

Galois theory

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ABSTRACT. The notes correspond to the bachelor course **Galois Theory** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences.

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Introduction

The notes correspond to the bachelor course **Galois theory** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Science. The course is divided into twelve two-hour lectures.

The material is somewhat standard. Basic texts on fields and Galois theory are for example [4] and [5].

As usual, we also mention a set of **great expository papers** by Keith Conrad. The notes are extremely well-written and useful at every stage of a mathematical career.

The notes include many exercises, some with detailed solutions. Mandatory exercises have a **green background**, while optional ones (bonus exercises) have a **yellow background**. The notes also include some additional comments. While these are entirely optional, I hope they offer further insight. They are highlighted with a **pink background**.

The notes include Magma code, which we use to verify examples and offer alternative solutions to certain exercises. Magma [1] is a powerful software tool designed for working with algebraic structures. There is a free **online** version of Magma available.

Several of the exercise solutions were written by Silvia Properzi.

These notes are freely available at <https://github.com/vendramin/galois>, where the most recent version can always be found. They are also listed in the AMS Open Math Notes collection, reference OMN:202508.111468; the copy hosted there is a snapshot and may lag behind the version on GitHub.

If you want to cite these notes, you may use the following entry:

```
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§ 1. Fields

Lecture 1

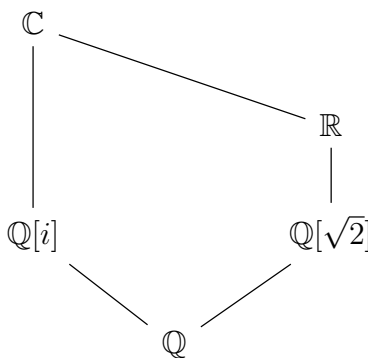
Throughout these notes, rings have an identity. Ring homomorphisms preserve the identity, and subrings contain the identity of the ambient ring.

Recall that a **field** is a commutative ring such that $1 \neq 0$ and every non-zero element is invertible. Examples of (infinite) fields are $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. If p is a prime number, then $\mathbb{Z}/p$ is a field.

1.1. EXAMPLE. The abelian group $\mathbb{Z}/2 \times \mathbb{Z}/2$ is a field with multiplication

$$(a, b)(c, d) = (ac + bd, ad + bc + bd).$$

1.2. EXAMPLE. $\mathbb{Q}[i] = \{a + bi : a, b \in \mathbb{Q}\}$ and $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ are fields.



1.3. EXERCISE. Prove that $\mathbb{Q}[i]$ and $\mathbb{Q}[\sqrt{2}]$ are not isomorphic as fields.

If R is a ring, there exists a unique ring homomorphism $\mathbb{Z} \rightarrow R$, $m \mapsto m1$. The image

$$R_{\mathbb{Z}} = \{m1 : m \in \mathbb{Z}\}$$

of this homomorphism is a subring of R , and it is known as the **prime subring** of R . The kernel is a subgroup of $\mathbb{Z}$ generated by some $t \geq 0$. The integer t is the **characteristic** of the ring R .

1.4. EXERCISE. The characteristic of a field is either zero or a prime number.

1.5. EXAMPLE. The characteristic of the field of Example 1.1 is two. Why?

Recall that a non-zero commutative ring R is an **integral domain** if $xy = 0 \implies x = 0$ or $y = 0$. Fields are integral domains.

1.6. EXERCISE. Let K be a field. Prove that the following statements are equivalent:

- 1) K is of characteristic zero.
- 2) The additive order of 1 is infinite.
- 3) The additive order of each $x \neq 0$ is infinite.
- 4) The prime subring of K is isomorphic to $\mathbb{Z}$.

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1.7. EXERCISE. Let p be a prime number and let K be a field. Prove that the following statements are equivalent:

- 1) K is of characteristic p .
- 2) The additive order of 1 is p .
- 3) The additive order of each $x \neq 0$ is p .
- 4) The prime subring of K is isomorphic to $\mathbb{Z}/p$.

Note that finite fields are of prime characteristic.

1.8. LEMMA. Let R be a commutative ring, $z \in R$ and $f(X) \in R[X]$ be a non-constant polynomial. Then z is a root of f if and only if $X - z$ divides f in $R[X]$.

In particular, if R is an integral domain, then f has at most $\deg(f)$ roots in R .

PROOF. Clearly, if $X - z$ divides f in $R[X]$, then z is a root of f .

Let $n = \deg(f)$ and $f = \sum_{i=0}^n a_i X^i$, for some $a_0, \dots, a_n \in R$. Then

$$f(X) - f(z) = \sum_{i=0}^n a_i X^i - \sum_{i=0}^n a_i z^i = \sum_{i=0}^n a_i (X^i - z^i) = \sum_{i=1}^n a_i (X^i - z^i).$$

Note that for every integer $i > 0$,

$$X^i - z^i = \sum_{k=0}^{i-1} z^k X^{i-k} - \sum_{k=0}^{i-1} z^{k+1} X^{i-1-k} = (X - z) \sum_{k=0}^{i-1} z^k X^{i-k-1},$$

hence $X - z$ divides $X^i - z^i$ in $R[X]$ for every $i > 0$. So $X - z$ divides $f(X) - f(z)$ in $R[X]$. If we now assume that z is a root of f , then $f(X) = f(X) - f(z)$ and it is divisible by $X - z$ in $R[X]$.

Suppose now that R is an integral domain. We prove the second part of the lemma by induction on the degree of f . If f has degree 1, then $f(X) = aX + b$ for some $a, b \in R$ with $a \neq 0$. Given two roots $z_1, z_2 \in R$ of f ,

$$az_1 = -b = az_2,$$

so $a(z_1 - z_2) = 0$. But since R is a domain and $a \neq 0$, necessarily $z_1 = z_2$. Assume now that $\deg(f) = n > 1$ and that every non-constant polynomial $g(X) \in R[X]$ of degree $< n$ has at most $\deg(g)$ roots in R . If f has no roots in R , then the result is immediate. If f has a root $z \in R$, then, by the previous part of this lemma, there exists $g(X) \in R[X]$ such that

$$f(X) = (X - z)g(X).$$

Hence, $0 < \deg(g) = \deg(f) - 1 = n - 1$. Therefore, we can apply the inductive hypothesis to g and obtain that g has at most $n - 1$ roots in R . To conclude, we can observe that $t \in R$ is a root of f if and only if $0 = f(t) = (t - z)g(t)$. Since R is a domain, the last equality is equivalent to $t = z$ or $g(t) = 0$. Thus f has at most n distinct roots. $\square$

Note that for the second part of the previous lemma, the hypothesis that R is an integral domain is necessary, as the following examples show.

1.9. EXAMPLE. The polynomial $X^2 - 1 \in (\mathbb{Z}/8)[X]$ has 4 roots in $\mathbb{Z}/8$: 1, 3, 5, and 7.

1.10. EXAMPLE. Let $R = H(\mathbb{R})$ be the ring of real quaternions. Then $X^2 + 1 \in R[X]$ has infinitely many roots in R .

§ 1. Fields

1.11. PROPOSITION. *Let K be a field. Then any finite subgroup of $K^\times = K \setminus \{0\}$ is cyclic.*

PROOF. Let G be a finite subgroup of $K^\times$. Let $x \in G$ have maximal order N . We will show that any element of G is a power of x . Let $y \in G$ and $n = |y|$.

We claim that n divides N . If not, there exists a prime number p and a power $q = p^{\alpha+1}$ of p such that $p^{\alpha+1} \mid n$ and $p^\alpha \mid N$, but $p^{\alpha+1} \nmid N$. Let $z = x^{p^\alpha} y^{n/q}$. Then $|x^{p^\alpha}| = N/p^\alpha$ and $|y^{n/q}| = q$. Since $\gcd(N/p^\alpha, q) = 1$ and G is abelian,

$$|z| = \text{lcm}\{N/p^\alpha, q\} = Np > N,$$

a contradiction.

Note that $X^n - 1 \in K[X]$ has at most n roots in K . Moreover, $x^{kN/n}$ for $k \in \{0, \dots, n-1\}$ are n distinct elements of K , that are roots of $X^n - 1$. Therefore

$$\{x^{kN/n} \mid k \in \{0, \dots, n-1\}\}$$

is the set of all roots of $X^n - 1 \in K[X]$ in K . Since y has order n , it has to be one of these roots. Thus $y = x^{kN/n}$ for some $k \in \{0, \dots, n-1\}$. $\square$

1.12. DEFINITION. A **subfield** of a ring R is a subring of R that is also a field.

Note that if K is a subfield of E , then the characteristic of K coincides with the characteristic of E . Moreover, if $K \rightarrow L$ is a field homomorphism, then K and L have the same characteristic.

1.13. EXERCISE. Let K be a field of characteristic $p > 0$. Prove that $K \rightarrow K, x \mapsto x^{p^n}$, is a field homomorphism for all $n \in \mathbb{Z}_{\geq 0}$.

1.14. DEFINITION. A field K is **prime** if there are no proper subfields of K .

Examples of prime fields are $\mathbb{Q}$ and $\mathbb{Z}/p$ for a prime number p .

1.15. PROPOSITION. *Let K be a field. The following statements hold:*

- 1) K has a unique subfield which is prime; it is known as the **prime subfield** of K .
- 2) The prime subfield of K is either isomorphic to $\mathbb{Q}$ if the characteristic of K is zero, or it is isomorphic to $\mathbb{Z}/p$ for some prime number p if the characteristic of K is p .

PROOF. To prove the first claim let L be the intersection of all the subfields of K . Then L is a subfield of K . If F is a subfield of L , then F is a subfield of K . Thus $L \subseteq F$ and hence $F = L$, which proves that L is prime. If L_1 is a subfield of K and L_1 is prime, then $L \subseteq L_1$ and hence $L = L_1$.

Let K_0 be the prime subfield of K . Suppose that K is of characteristic $p > 0$. Then the prime subring $K_{\mathbb{Z}}$ of K is a field isomorphic to $\mathbb{Z}/p$ and hence $K_0 = K_{\mathbb{Z}} \simeq \mathbb{Z}/p$. Suppose now that the characteristic of K is zero. Let $E = \{m1(n1)^{-1} : m, n \in \mathbb{Z}, n \neq 0\}$. The map $\mathbb{Q} \rightarrow E, m/n \mapsto m1(n1)^{-1}$, is a field isomorphism. We claim that $K_0 = E$. Since $K_{\mathbb{Z}} \subseteq K_0$, it follows that $E \subseteq K_0$. Hence $E = K_0$, as E is a subfield of K . $\square$

1.16. DEFINITION. Let E be a field and K be a subfield of E . Then E is a **field extension** of K . We will use the notation E/K .

Let E/K be a field extension. Then E is a K -vector space with the usual scalar multiplication $K \times E \rightarrow E, (\lambda, x) \mapsto \lambda x$.

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1.17. DEFINITION. The **degree** of an extension E of K is the dimension $\dim_K E$, which may be infinite. It will be denoted by $[E : K]$.

We say that E is a **finite extension** of K if $[E : K]$ is finite.

1.18. EXAMPLE. Let K be a field. Then $[K : K] = 1$. Conversely, if E is an extension of K and $[E : K] = 1$, then $K = E$. If not, let $x \in E \setminus K$. We claim that $\{1, x\}$ is linearly independent over K . Indeed, if $a1 + bx = 0$ for some $a, b \in K$, then $bx = -a$. If $b \neq 0$, then $x = -a/b \in K$, a contradiction. If $b = 0$, then $a = 0$.

We know that $[\mathbb{C} : \mathbb{R}] = 2$.

1.19. EXAMPLE. A basis of $\mathbb{Q}[\sqrt{2}]$ over $\mathbb{Q}$ is given by $\{1, \sqrt{2}\}$. Then $[\mathbb{Q}[\sqrt{2}] : \mathbb{Q}] = 2$. The calculations can be easily done with Magma:

```
> Q<z> := QuadraticField(2);
> Characteristic(Q);
0
> K := PrimeField(Q);
> K;
Rational Field
> Degree(Q);
2
> Basis(Q);
[ 1, z ]
> z^2;
2
> One(K) eq One(Q);
true
> G := GaloisGroup(Q);
> G;
Symmetric group G acting on a set of cardinality 2
Order = 2
```

1.20. EXAMPLE. Since $\mathbb{Q}$ is countable and $\mathbb{R}$ is not, $[\mathbb{R} : \mathbb{Q}] > \aleph_0$. If $\{x_i : i \in \mathbb{Z}_{>0}\}$ is a countable basis of $\mathbb{R}$ over $\mathbb{Q}$, for each n consider the $\mathbb{Q}$ -vector space V_n generated by $\{x_1, \dots, x_n\}$. Each V_n is countable, hence

$$\mathbb{R} = \bigcup_{n \geq 1} V_n$$

is countable, a contradiction.

If E is an extension of K and E is finite as a set, then $[E : K]$ is finite.

1.21. PROPOSITION. *Let K be a finite field. Then $|K| = p^m$ for some prime number p and some $m \geq 1$.*

PROOF. We know the prime subfield K_0 of K is isomorphic to $\mathbb{Z}/p$. In particular, $|K_0| = p$. Since K is finite, $[K : K_0] = m$ for some m . If $\{x_1, \dots, x_m\}$ is a basis of K over K_0 , then each element of K can be written uniquely as $\sum_{i=1}^m a_i x_i$ for some $a_1, \dots, a_m \in K_0$. Then there is a bijection between K and K_0^m and hence $|K| = |K_0^m| = p^m$. $\square$

§ 1. Fields

We now perform some basic calculations with a finite field of eight elements:

```
> E<x> := FiniteField(8);
> PrimeField(E);
Finite field of size 2
> Degree(E);
3
> Characteristic(E);
2
> #E;
8
> { x : x in E };
{ 1, x, x^2, x^3, x^4, x^5, x^6, 0 }
```

Here is an alternative construction of a field of eight elements as a quotient of a polynomial ring:

```
> R<x> := PolynomialRing(PrimeField(E));
> IsIrreducible(x^3+x+1);
true
> I := ideal< R | x^3+x+1 >;
> IsMaximal(I);
true
> Q<y> := quo< R | I >;
> IsField(Q);
true
> { y : y in Q };
{
  y^2 + 1,
  0,
  y^2,
  y + 1,
  1,
  y,
  y^2 + y + 1,
  y^2 + y
}
> #Q;
8
```

1.22. DEFINITION. Let E be an extension of K . A **subextension** F/K of E/K is a subfield F of E that contains K , that is $K \subseteq F \subseteq E$.

1.23. DEFINITION. Let E and E_1 be extensions over K . An **extension homomorphism**

$$E/K \rightarrow E_1/K$$

is a field homomorphism $\sigma: E \rightarrow E_1$ such that $\sigma(x) = x$ for all $x \in K$.

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To describe the homomorphism $\sigma: E/K \rightarrow E_1/K$ of the extensions over K one typically writes the commutative diagram

$$\begin{array}{ccc} K & \xlongequal{\quad} & K \\ \downarrow & & \downarrow \\ E & \xrightarrow{\sigma} & E_1 \end{array}$$

We write $\text{Hom}(E/K, E_1/K)$ to denote the set of homomorphisms $E/K \rightarrow E_1/K$ of extensions of K . Note that if $\sigma \in \text{Hom}(E/K, E_1/K)$, then σ is a K -linear map, as

$$\sigma(\lambda x) = \sigma(\lambda)\sigma(x) = \lambda\sigma(x)$$

for all $\lambda \in K$ and $x \in E$.

1.24. EXAMPLE. The conjugation map $\mathbb{C} \rightarrow \mathbb{C}$, $z \mapsto \bar{z}$, is an endomorphism of $\mathbb{C}$ as an extension over $\mathbb{R}$. Let $\varphi \in \text{Hom}(\mathbb{C}/\mathbb{R}, \mathbb{C}/\mathbb{R})$. Then

$$\varphi(x + iy) = \varphi(x) + \varphi(i)\varphi(y) = x + \varphi(i)y$$

for all $x, y \in \mathbb{R}$. Since $\varphi(i)^2 = \varphi(i^2) = \varphi(-1) = -1$, it follows that $\varphi(i) \in \{-i, i\}$. Thus either $\varphi(x + iy) = x + iy$ or $\varphi(x + iy) = x - iy$.

1.25. EXERCISE. Let K be a field, K_0 be its prime subfield and $\sigma: K \rightarrow K$ be a field homomorphism. Prove that $\sigma \in \text{Hom}(K/K_0, K/K_0)$.

If E/K is an extension, then

$$\begin{aligned} \text{Aut}(E/K) &= \{\sigma: E/K \rightarrow E/K \text{ is a bijective extension homomorphism}\} \\ &= \{\sigma: E \rightarrow E : \sigma \text{ is a bijective field homomorphism with } \sigma|_K = \text{id}\} \end{aligned}$$

is a group with composition.

1.26. DEFINITION. Let E/K be an extension. The **Galois group** of E/K is the group $\text{Aut}(E/K)$ and it will be denoted by $\text{Gal}(E/K)$.

A typical example: $\text{Gal}(\mathbb{C}/\mathbb{R}) \simeq \mathbb{Z}/2$.

As an example, we show with the computer that $\text{Gal}(\mathbb{Q}[\sqrt{2}]/\mathbb{Q}) \simeq \mathbb{Z}/2$:

```
> Q<z> := QuadraticField(2);
> G := GaloisGroup(Q);
> G;
Symmetric group G acting on a set of cardinality 2
Order = 2
> GroupName(G);
C2
```

1.27. EXERCISE. Prove that the polynomial $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$.

The previous exercise can easily be solved using Magma:

```
> Q<x> := PolynomialRing(Rationals());
> f := x^3-2;
> IsIrreducible(f);
true
```

§ 1. Fields

1.28. EXAMPLE. Let $\theta = \sqrt[3]{2}$ and let $E = \{a + b\theta + c\theta^2 : a, b, c \in \mathbb{Q}\}$. Note that

$$a + b\theta + c\theta^2 = 0 \iff a = b = c = 0.$$

Since $X^3 - 2$ is irreducible in $\mathbb{Q}[X]$, evaluation at θ yields $E \simeq \mathbb{Q}[X]/(X^3 - 2)$. Then E is an extension of $\mathbb{Q}$ such that $[E : \mathbb{Q}] = 3$. We claim that $\text{Gal}(E/\mathbb{Q})$ is trivial. If $\sigma \in \text{Gal}(E/\mathbb{Q})$ and $z = a + b\theta + c\theta^2$, then $\sigma(z) = a + b\sigma(\theta) + c\sigma(\theta)^2$. Since $\sigma(\theta) \in E \subseteq \mathbb{R}$ and

$$\sigma(\theta)^3 = \sigma(\theta^3) = \sigma(2) = 2,$$

it follows that $\sigma(\theta) = \theta$ and therefore $\sigma = \text{id}$.

Recall that we have the following characterization of irreducible polynomials, known as **Gauss' Lemma**.

1.29. LEMMA (Gauss' lemma). *Let A be a unique factorization domain and K be its fraction field. A non-constant polynomial $f \in A[X]$ is irreducible in $A[X]$ if and only if it is primitive and irreducible in $K[X]$.*

In particular, a non-constant polynomial $f \in \mathbb{Z}[X]$ is irreducible if and only if it is primitive and irreducible in $\mathbb{Q}[X]$. This helps with the following exercise, which is known as **Eisenstein's irreducibility criterion**:

1.30. EXERCISE. Let $f = \sum_{i=0}^n a_i X^i \in \mathbb{Z}[X]$ be a polynomial of degree $n > 0$. Assume that there exists a prime number p such that $p \mid a_i$ for all $i \in \{0, 1, \dots, n-1\}$, $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible in $\mathbb{Q}[X]$.

1.31. BONUS EXERCISE. Let A be a unique factorization domain and K be its fraction field. Let $f = \sum_{i=0}^n a_i X^i \in A[X]$ be a polynomial of degree $n > 0$. Assume that there exists a prime element $p \in A$ such that $p \mid a_i$ for all $i \in \{0, 1, \dots, n-1\}$, $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible in $K[X]$.

1.32. EXERCISE. Prove that the polynomials

$$\begin{aligned} f &= X^{10} + 60X^7 + 82X^6 - 36X^3 + 2, \\ g &= 3X^{10} + 15X^2 - 45, \end{aligned}$$

are irreducible in $\mathbb{Q}[X]$.

The previous exercise is easy for Magma:

```
> R<x> := PolynomialRing(Rationals());
> g := 3*x^10+15*x^2-45;
> IsIrreducible(g);
true
> f := x^10+60*x^7+82*x^6-36*x^3+2;
> IsIrreducible(f);
true
```

1.33. EXERCISE. Is the polynomial $f = 3(X^{10} + 5X^2 - 15)$ irreducible in $\mathbb{Z}[X]$?

Let us see what Magma can do here:

§ 2. Algebraic extensions

```
> R<x> := PolynomialRing(Integers());
> f := 3*x^10+15*x^2-45;
> IsIrreducible(f);
false
> Factorization(f);
[
  <3, 1>,
  <x^10 + 5*x^2 - 15, 1>
]
```

If E/K is an extension and S is a subset of E , then there exists a unique smallest subextension F/K of E/K such that $S \subseteq F$. In fact,

$$F = \bigcap \{T : T \text{ is a subfield of } E \text{ that contains } K \cup S\}.$$

If L/K is a subextension of E/K such that $S \subseteq L$, then $F \subseteq L$ by definition. The extension F is known as the **subextension generated by S** and it will be denoted by $K(S)$. If $S = \{x_1, \dots, x_n\}$ is finite, then the extension $K(S)/K$, where $K(S) = K(x_1, \dots, x_n)$, is said to be of **finite type**.

1.34. EXAMPLE. If $\{e_1, \dots, e_n\}$ is a basis of E over K , then $E = K(e_1, \dots, e_n)$.

1.35. EXAMPLE. The field $\mathbb{Q}(\sqrt{2})$ is precisely the subextension of $\mathbb{R}/\mathbb{Q}$ generated by $\sqrt{2}$.

Let E/K be an extension and S and T be subsets of E . Then

$$K(S \cup T) = K(S)(T) = K(T)(S).$$

If, moreover, $S \subseteq T$, then $K(S) \subseteq K(T)$.

§ 2. Algebraic extensions

2.1. DEFINITION. Let E/K be an extension. An element $x \in E$ is **algebraic** over K if there exists a non-zero polynomial $f(X) \in K[X]$ such that $f(x) = 0$. If x is not algebraic over K , then it is called **transcendental** over K .

2.2. DEFINITION. An extension E/K is **algebraic** if every $x \in E$ is algebraic over K .

If K is a field, every $x \in K$ is algebraic over K , as x is a root of $X - x \in K[X]$. In particular, K/K is an algebraic extension.

2.3. EXAMPLE. $\mathbb{C}/\mathbb{R}$ is an algebraic extension. If $z \in \mathbb{C} \setminus \mathbb{R}$, then z is a root of the polynomial $X^2 - (z + \bar{z})X + |z|^2 \in \mathbb{R}[X]$.

If F/K is an extension and $x \in E$ is algebraic over K for some field $E \supseteq F$, then x is algebraic over F .

2.4. EXAMPLE. $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is algebraic, as the number $a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$ is a root of the polynomial $X^2 - 2aX + (a^2 - 2b^2) \in \mathbb{Q}[X]$.

The extension $\mathbb{C}/\mathbb{Q}$ is not algebraic. For example, Hermite proved that e is transcendental over $\mathbb{Q}$; see [5, Theorem 24.4]. Lindemann's theorem states that π is not algebraic over $\mathbb{Q}$; see [5, Theorem 24.5].

§ 2. Algebraic extensions

2.5. EXAMPLE. Let $a = \sqrt{2}$ and $b = \sqrt[3]{3}$. Both a and b are algebraic numbers over $\mathbb{Q}$. Let us show that $a + b$ is also algebraic. Let $f(X) = X^3 - 3 \in \mathbb{Q}[X]$. Then $f(b) = 0$. Note that the polynomial

$$g(X) = f(X - a) = X^3 - 3aX^2 + 3a^2X - a^3 - 3 \in \mathbb{Q}(a)[X]$$

is such that $g(a + b) = 0$. How can we find a polynomial with coefficients in $\mathbb{Q}$ that vanishes on $a + b$? We do the “conjugation” trick:

$$h(X) = f(X - a)f(X + a) = X^6 - 6X^4 - 6X^3 + 12X^2 - 36X + 1 \in \mathbb{Q}[X].$$

Note that $h(a + b) = 0$. How can you prove that ab is also algebraic over $\mathbb{Q}$?

If E/K is an extension and $x \in E$ is algebraic over K , then the evaluation homomorphism $K[X] \rightarrow E, p \mapsto p(x)$, is not injective. In particular, its kernel is a non-zero ideal. Hence it is generated by a monic polynomial f .

2.6. DEFINITION. Let E/K be an extension and $x \in E$ be an algebraic element. The monic polynomial that generates the kernel of $K[X] \rightarrow E, f \mapsto f(x)$, is known as the **minimal polynomial** of x over K and it will be denoted by $f(x, K)$. The **degree** of x over K is then $\deg f(x, K)$.

Some basic properties of the minimal polynomial of an algebraic element:

2.7. PROPOSITION. Let E/K be an extension and $x \in E$. Assume that x is algebraic over K .

- 1) If $g \in K[X]$ is non-zero and such that $g(x) = 0$, then $f(x, K)$ divides g and $\deg f(x, K) \leq \deg g$.
- 2) $f(x, K)$ is irreducible in $K[X]$.
- 3) If F/K is a subextension of E/K , then $f(x, F)$ divides $f(x, K)$.

PROOF. Write $f = f(x, K)$ to denote the minimal polynomial of x . To prove 1) note that $g(x) = 0$ implies that g belongs to the kernel of the evaluation map, so g is a multiple of f . To prove 2) note that if $f = pq$ for some $p, q \in K[X]$ such that $0 < \deg p, \deg q < \deg f$, then $f(x) = 0$ implies that either $p(x) = 0$ or $q(x) = 0$, a contradiction. Finally, we prove 3). Since $f \in K[X] \subseteq F[X]$ and $f(x) = 0$, it follows from 1) that $f(x, F)$ divides f . $\square$

Some easy examples: $f(i, \mathbb{Q}) = X^2 + 1$, $f(i, \mathbb{C}) = X - i$ and $f(\sqrt[3]{2}, \mathbb{Q}) = X^3 - 2$. Here is the code:

```
> R<x> := PolynomialRing(Rationals());
> Q<i> := QuadraticField(-1);
> MinimalPolynomial(i);
x^2 + 1
```

A bit harder: $f(\sqrt[3]{2} + 1, \mathbb{Q}) = X^3 - 3X^2 + 3X - 3$. Here is the code:

```
> R<x> := PolynomialRing(Rationals());
> E<z> := NumberField(x^3-2);
> MinimalPolynomial(z+1);
x^3 - 3*x^2 + 3*x - 3
```

§ 2. Algebraic extensions

2.8. EXAMPLE. Let us compute $f(\sqrt{2} + \sqrt{3}, \mathbb{Q})$. Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\begin{aligned}\alpha - \sqrt{2} = \sqrt{3} &\implies (\alpha - \sqrt{2})^2 = 3 \implies \alpha^2 - 2\sqrt{2}\alpha + 2 = 3 \\ &\implies \alpha^2 - 1 = 2\sqrt{2}\alpha \implies (\alpha^2 - 1)^2 = 8\alpha^2 \implies \alpha^4 - 10\alpha^2 + 1 = 0.\end{aligned}$$

Thus α is a root of $g = X^4 - 10X^2 + 1$. To prove that $g = f(\alpha, \mathbb{Q})$ it is enough to prove that g is irreducible in $\mathbb{Q}[X]$. First note that the roots of g are $\sqrt{2} + \sqrt{3}$, $\sqrt{2} - \sqrt{3}$, $-\sqrt{2} + \sqrt{3}$ and $-\sqrt{2} - \sqrt{3}$. This means that if g is not irreducible, then $g = hh_1$ for some polynomials $h, h_1 \in \mathbb{Q}[X]$ such that $\deg h = \deg h_1 = 2$. This is not possible, as

$$\begin{aligned}(\sqrt{2} + \sqrt{3}) + (\sqrt{2} - \sqrt{3}) &= 2\sqrt{2} \notin \mathbb{Q}, \\ (\sqrt{2} + \sqrt{3}) + (-\sqrt{2} + \sqrt{3}) &= 2\sqrt{3} \notin \mathbb{Q}, \\ (\sqrt{2} + \sqrt{3})(-\sqrt{2} - \sqrt{3}) &= -5 - 2\sqrt{6} \notin \mathbb{Q}.\end{aligned}$$

2.9. PROPOSITION. Let F/K be a subextension of E/K . Then

$$[E : K] = [E : F][F : K].$$

PROOF. Let $\{e_i : i \in I\}$ be a basis of E over F and $\{f_j : j \in J\}$ be a basis of F over K . If $x \in E$, then $x = \sum_i \lambda_i e_i$ (finite sum) for some $\lambda_i \in F$. For each i , $\lambda_i = \sum_j a_{ij} f_j$ (finite sum) for some $a_{ij} \in K$. Then $x = \sum_i \sum_j a_{ij} (f_j e_i)$. This means that $\{f_j e_i : i \in I, j \in J\}$ generates E as a K -vector space. Let us prove that $\{f_j e_i : i \in I, j \in J\}$ is linearly independent. If $\sum_i \sum_j a_{ij} (f_j e_i) = 0$ (finite sum) for some $a_{ij} \in K$, then

$$\begin{aligned}0 = \sum_i \left(\sum_j a_{ij} f_j \right) e_i &\implies \sum_j a_{ij} f_j = 0 \text{ for all } i \in I \\ &\implies a_{ij} = 0 \text{ for all } i \in I \text{ and } j \in J. \quad \square\end{aligned}$$

We state a lemma:

2.10. LEMMA. If A is a finite-dimensional commutative algebra over K and A is an integral domain, then A is a field.

PROOF. Let $a \in A \setminus \{0\}$. We need to prove that there exists $b \in A$ such that $ab = 1$. Let $\theta : A \rightarrow A$, $x \mapsto ax$. Note that θ is a K -linear transformation, as

$$\theta(x + y) = a(x + y) = ax + ay = \theta(x) + \theta(y), \quad \theta(\lambda x) = a(\lambda x) = \lambda(ax) = \lambda\theta(x),$$

for all $x, y \in A$ and $\lambda \in K$. It is injective since A is an integral domain. Since $\dim_K A < \infty$, it follows that θ is an isomorphism. In particular, $\theta(A) = A$, which implies that there exists $b \in A$ such that $1 = ab$. $\square$

Let E/K be an extension and $x \in E$. Then

$$K[x] = \{f(x) : f \in K[X]\}$$

is a subring of E that contains K . Note that $K[x]$ is a K -vector space.

More generally, if $x_1, \dots, x_n \in E$, then

$$K[x_1, \dots, x_n] = \{f(x_1, \dots, x_n) : f \in K[X_1, \dots, X_n]\}$$

§ 2. Algebraic extensions

is a subring of E . Note that $K[x_1, \dots, x_n]$ is a K -vector space. Clearly, $K[x_1, \dots, x_n]$ is a domain and

$$K(x_1, \dots, x_n) = \left\{ \frac{f(x_1, \dots, x_n)}{g(x_1, \dots, x_n)} : f, g \in K[X_1, \dots, X_n] \text{ with } g(x_1, \dots, x_n) \neq 0 \right\}$$

is the extension of K generated by $x_1, \dots, x_n$. Note that

$$K(x_1, \dots, x_n) = (K(x_1, \dots, x_{n-1}))(x_n).$$

The previous construction can be generalized. Let I be a non-empty set. For each $i \in I$, let X_i be a variable. Consider the polynomial ring $K[\{X_i : i \in I\}]$ and let $S = \{x_i : i \in I\}$ be a subset of E . There exists a unique algebra homomorphism

$$K[\{X_i : i \in I\}] \rightarrow E$$

such that $X_i \mapsto x_i$ for all $i \in I$. The image is denoted by $K[S]$. In particular, an element $z \in K[S]$ is of the form

$$z = h(x_1, \dots, x_n)$$

for a polynomial $h \in K[X_1, \dots, X_n]$ in finitely many variables $X_1, \dots, X_n$ and $x_1, \dots, x_n \in S$.

2.11. EXERCISE. Prove that $\mathbb{Q}[\sqrt{2}] = \mathbb{Q}(\sqrt{2})$.

The exercise is not an accident.

2.12. THEOREM. Let E/K be an extension and $x \in E \setminus K$. The following statements are equivalent:

- 1) x is algebraic over K .
- 2) $\dim_K K[x] < \infty$.
- 3) $K[x]$ is a field.
- 4) $K[x] = K(x)$.

PROOF. We first prove 1) $\implies$ 2). Let $z \in K[x]$, say $z = h(x)$ for some $h \in K[X]$. There exists $g \in K[X]$ such that $g \neq 0$ and $g(x) = 0$. Divide h by g to obtain polynomials $q, r \in K[X]$ such that $h = gq + r$, where $r = 0$ or $\deg r < \deg g$. This implies that

$$z = h(x) = g(x)q(x) + r(x) = r(x).$$

If $\deg g = m$, then $r = \sum_{i=0}^{m-1} a_i X^i$ for some $a_0, \dots, a_{m-1} \in K$. Thus

$$z = \sum_{i=0}^{m-1} a_i x^i$$

and hence $K[x] \subseteq \langle 1, x, \dots, x^{m-1} \rangle$.

The previous lemma proves that 2) $\implies$ 3).

It is trivial that 3) $\implies$ 4).

It remains to prove that 4) $\implies$ 1). Since $x \neq 0$, $1/x \in K(x) = K[x]$. There exists $a_0, \dots, a_n \in K$ such that $1/x = a_0 + a_1 x + \dots + a_n x^n$. Thus

$$a_n x^{n+1} + \dots + a_1 x^2 + a_0 x - 1 = 0,$$

and hence x is a root of $a_n X^{n+1} + \dots + a_0 X - 1 \in K[X] \setminus \{0\}$. □

§ 2. Algebraic extensions

Note that if x is algebraic over K , then $K[x] \simeq K[X]/(f(x, K))$.

2.13. EXERCISE. Let E/K be an extension and $x \in E$ be an algebraic element over K . Prove that the degree of x over K is equal to $[K(x) : K]$.

2.14. COROLLARY. If E/K is finite, then E/K is algebraic.

PROOF. Let $n = [E : K]$ and $x \in E \setminus K$. The set $\{1, x, \dots, x^n\}$ has $n + 1$ elements, so it is linearly dependent. There exist $a_0, \dots, a_n \in K$, not all zero, such that

$$a_0 + a_1x + \dots + a_nx^n = 0.$$

Thus x is a root of the non-zero polynomial $a_0 + a_1X + \dots + a_nX^n \in K[X]$. $\square$

In Example 2.5 we proved that $\sqrt{2} + \sqrt[3]{3}$ and $\sqrt{2}\sqrt[3]{3}$ are algebraic over $\mathbb{Q}$. This can be easily proved now with Corollary 2.14.

2.15. EXERCISE. Let E/K be an extension and a and b be algebraic over K . Prove that $a + b$ and ab are algebraic over K .

We note that the converse of Corollary 2.14 does not hold.

2.16. COROLLARY. If E/K is an extension and $x_1, \dots, x_n \in E$ are algebraic over K , then $K(x_1, \dots, x_n)/K$ is finite and $K(x_1, \dots, x_n) = K[x_1, \dots, x_n]$.

PROOF. We proceed by induction on n . The case $n = 1$ follows immediately from the theorem. So assume the result holds for some $n \geq 1$. Since the extensions

$$K(x_1, \dots, x_n)/K(x_1, \dots, x_{n-1}) \quad \text{and} \quad K(x_1, \dots, x_{n-1})/K$$

are both finite, it follows that $K(x_1, \dots, x_n)/K$ is finite. Moreover,

$$\begin{aligned} K(x_1, \dots, x_n) &= K(x_1, \dots, x_{n-1})(x_n) \\ &= K(x_1, \dots, x_{n-1})[x_n] = K[x_1, \dots, x_{n-1}][x_n] = K[x_1, \dots, x_n]. \end{aligned} \quad \square$$

2.17. COROLLARY. Let $E = K(S)$ for some set S . Then E/K is algebraic if and only if x is algebraic over K for all $x \in S$.

PROOF. Let us prove the non-trivial implication. Let $z \in K(S)$. In particular, there exists a finite subset $T \subseteq S$ such that $z \in K(T)$. The previous result implies that $K(T)/K$ is algebraic, and hence z is algebraic. $\square$

If E/K is an extension, let

$$\overline{K}_E = \{x \in E : x \text{ is algebraic over } K\}.$$

2.18. COROLLARY. If E/K is an extension, then $\overline{K}_E$ is a subfield of E that contains K . Moreover, $K(\overline{K}_E) = \overline{K}_E$ and $K(\overline{K}_E)/K$ is algebraic.

PROOF. By definition, $K(\overline{K}_E)/K$ is algebraic. Thus $K(\overline{K}_E) \subseteq \overline{K}_E$. From this, it follows that $K(\overline{K}_E) = \overline{K}_E$. $\square$

The following exercise is now almost trivial:

2.19. EXERCISE. Let E/K be an extension of finite type; this means that $E = K(S)$ for some finite set S . Prove that E/K is algebraic if and only if E/K is finite.

§ 2. Algebraic extensions

Let $\overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} : \alpha \text{ is algebraic over } \mathbb{Q}\}$. Then $\overline{\mathbb{Q}}$ is the field of algebraic numbers. Can you compute $[\overline{\mathbb{Q}} : \mathbb{Q}]$?

2.20. EXERCISE. Prove that $[\mathbb{Q}[\sqrt[3]{2}] : \mathbb{Q}] = 3$.

For the previous exercise, you may use Eisenstein's criterion.

2.21. EXERCISE. Let $E = \mathbb{Q}[i, \sqrt{2}] = \mathbb{Q}[\sqrt{2}][i]$. Prove that $[E : \mathbb{Q}] = 4$.

2.22. EXERCISE. Let $E = \mathbb{Q}[\sqrt{2}, \sqrt[3]{5}]$.

- 1) Compute $[E : \mathbb{Q}]$.
- 2) Prove that $E = \mathbb{Q}[\sqrt{2} + \sqrt[3]{5}]$.
- 3) Find the minimal polynomial of $\sqrt{2} + \sqrt[3]{5}$ over $\mathbb{Q}$.

2.23. EXERCISE. Find the minimal polynomials of $\sqrt[4]{3}i$ over $\mathbb{Q}[i]$ and over $\mathbb{Q}[\sqrt{3}]$.

2.24. EXERCISE. Find the minimal polynomial of $\sqrt{2} + \sqrt[3]{5}i$ over $\mathbb{Q}[i]$.

Algebraic field extensions form a nice class of extensions. The same happens with finite field extensions.

2.25. PROPOSITION. *Let F/K be a subextension of E/K . Then E/K is algebraic if and only if E/F and F/K are algebraic.*

PROOF. If E/K is algebraic, then E/F and F/K are both algebraic, as $K \subseteq F \subseteq E$. Let us assume that E/F and F/K are both algebraic. Let $x \in E$ and let L be the subextension over K generated by the coefficients of $f(x, F)$, the minimal polynomial of x over F . Then L/K is finite, since it is generated by finitely many algebraic elements. Moreover, x is algebraic over L . Since

$$[L(x) : K] = [L(x) : L][L : K] < \infty,$$

$L(x)/K$ is algebraic. In particular, x is algebraic over K . □

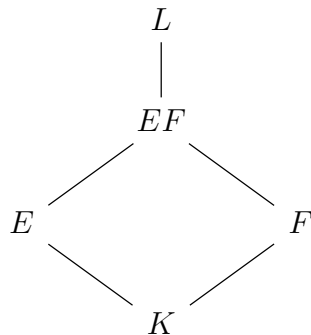
2.26. EXERCISE. Let F/K be a subextension of E/K . Prove that E/K is finite if and only if E/F and F/K are finite.

Let K be a field and $K \subseteq F \subseteq L$ and $K \subseteq E \subseteq L$ be fields. The **composite** of E and F is defined as

$$EF = K(E \cup F) = F(E) = E(F)$$

§ 2. Algebraic extensions

and it is equal to the smallest field that contains E and F . Here is the picture:



2.27. EXERCISE. Let E/K and F/K be algebraic field extensions. Prove that

$$EF = \left\{ \sum_{i=1}^m e_i f_i : m \in \mathbb{Z}_{>0}, e_i \in E, f_i \in F \text{ for all } i \in \{1, \dots, m\} \right\}.$$

2.28. EXERCISE. If $E = \mathbb{Q}(\sqrt{2})$ and $F = \mathbb{Q}(\sqrt{3})$, then $EF = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$ and $\mathbb{Q}(\sqrt{2}) \cap \mathbb{Q}(\sqrt{3})$.

2.29. EXERCISE. Let $\xi \in \mathbb{C}$ be a primitive cubic root of one. If $E = \mathbb{Q}(\sqrt[3]{2})$ and $F = \mathbb{Q}(\xi)$, then $EF = \mathbb{Q}(\sqrt[3]{2}, \xi)$. Compute $[\mathbb{Q}(\sqrt[3]{2}, \xi) : \mathbb{Q}]$ and $\mathbb{Q}(\sqrt[3]{2}) \cap \mathbb{Q}(\xi)$.

2.30. EXERCISE. Let E/K and F/K be extensions, where both E and F are subfields of a field L . If F/K is algebraic, then EF/E is algebraic.

2.31. EXERCISE. Let E/K and F/K be extensions, where both E and F are subfields of a field L . If F/K is finite, then EF/E is finite.

The solution to the previous exercise shows, in particular, that $[EF : E] \leq [F : K]$.

2.32. LEMMA. Let $\sigma : K \rightarrow L$ be a field homomorphism. Then there exists an extension E/K and a field isomorphism $\varphi : E \rightarrow L$ such that $\varphi|_K = \sigma$.

PROOF. Note that $\sigma : K \rightarrow \sigma(K)$ is bijective. Let A be a set in bijection with $L \setminus \sigma(K)$ and disjoint with K . Let $E = K \cup A$. If $\theta : A \rightarrow L \setminus \sigma(K)$ is bijective, then let

$$\varphi : E \rightarrow L, \quad \varphi(x) = \begin{cases} \sigma(x) & \text{if } x \in K, \\ \theta(x) & \text{if } x \in A. \end{cases}$$

Then φ is a bijective map such that $\varphi|_K = \sigma$. Transport the operations of L onto E , that is to define binary operations on E as follows:

$$(x, y) \mapsto x \oplus y = \varphi^{-1}(\varphi(x) + \varphi(y)), \quad (x, y) \mapsto x \odot y = \varphi^{-1}(\varphi(x)\varphi(y)).$$

Then, for example,

$$x \oplus y = \varphi^{-1}(\varphi(x) + \varphi(y)) = \varphi^{-1}(\sigma(x) + \sigma(y)) = \varphi^{-1}(\sigma(x + y)) = \varphi^{-1}(\varphi(x + y)) = x + y$$

for all $x, y \in K$. □

§ 2. Algebraic extensions

If $\sigma: A \rightarrow B$ is a ring homomorphism, then σ induces a ring homomorphism

$$\bar{\sigma}: A[X] \rightarrow B[X], \quad \sum_i a_i X^i \mapsto \sum_i \sigma(a_i) X^i.$$

2.33. THEOREM. *Let K be a field and $f \in K[X]$ be such that $\deg f > 0$. Then there exists an extension E/K such that f admits a root in E .*

PROOF. We may assume that f is irreducible over K . Let $L = K[X]/(f)$ and

$$\pi: K[X] \rightarrow L$$

be the canonical map. Then L is a field (the reader should explain why). Let $\sigma: K \rightarrow L$, $a \mapsto \pi(aX^0)$, and $g = \bar{\sigma}(f) \in L[X]$.

We claim that $\pi(X)$ is a root of g in L . Suppose that $f = \sum_i a_i X^i$. Then

$$\begin{aligned} g(\pi(X)) &= \bar{\sigma}(f)(\pi(X)) \\ &= \sum_i \sigma(a_i) \pi(X)^i = \sum_i \pi(a_i X^0) \pi(X^i) = \pi\left(\sum_i a_i X^i\right) = \pi(f) = 0. \end{aligned}$$

By Lemma 2.32, there exists an extension E/K and an isomorphism $\varphi: E \rightarrow L$ such that $\varphi|_K = \sigma$. Note that $\varphi(x) = 0$ if and only if $x = 0$. If $u = \pi(X)$, then $\varphi^{-1}(u)$ is a root of f in E , as

$$\begin{aligned} \varphi(f(\varphi^{-1}(u))) &= \varphi\left(\sum_i a_i \varphi^{-1}(u)^i\right) = \varphi\left(\sum_i a_i \varphi^{-1}(u^i)\right) \\ &= \sum_i \varphi(a_i) u^i = \sum_i \sigma(a_i) u^i = g(u) = 0. \end{aligned} \quad \square$$

As a corollary, if K is a field and $f_1, \dots, f_n \in K[X]$ are polynomials of positive degree, then there exists an extension E/K such that each f_i admits a root in E . This is proved by induction on n .

2.34. DEFINITION. A field K is **algebraically closed** if each $f \in K[X]$ of positive degree admits a root in K .

The **fundamental theorem of algebra** states that $\mathbb{C}$ is algebraically closed. A typical proof uses complex analysis. Later we will give a proof of this result using Galois theory.

2.35. PROPOSITION. *The following statements are equivalent:*

- 1) K is algebraically closed.
- 2) If $f \in K[X]$ is irreducible, then $\deg f = 1$.
- 3) If $f \in K[X]$ is non-zero, then f decomposes linearly in $K[X]$, that is

$$f = a \prod_{i=1}^n (X - \alpha_i)^{m_i}$$

for some $a \in K$ and $\alpha_1, \dots, \alpha_n \in K$.

- 4) If E/K is algebraic, then $E = K$.

PROOF. 1) $\implies$ 2) $\implies$ 3) are exercises.

§ 3. Artin's theorem

Let us prove that 3) $\implies$ 4). Let $x \in E$. Decompose $f(x, K)$ linearly in $K[X]$ as

$$f(x, K) = a \prod_{i=1}^n (X - \alpha_i)^{m_i}$$

and evaluate on x to obtain that $x = \alpha_j$ for some j .

To prove that 4) $\implies$ 1) let $f \in K[X]$ be such that $\deg f > 0$. There exists an extension E/K such that f has a root x in E . The extension $K(x)/K$ is algebraic and hence $K(x) = K$, so $x \in K$. $\square$

§ 3. Artin's theorem

3.1. DEFINITION. An **algebraic closure** of a field K is an algebraic extension C/K such that C is algebraically closed.

For example, $\mathbb{C}/\mathbb{R}$ is an algebraic closure but $\mathbb{C}/\mathbb{Q}$ is not.

3.2. PROPOSITION. *Let C be algebraically closed and $\sigma: K \rightarrow C$ be a field homomorphism. If E/K is algebraic, then there exists a field homomorphism $\varphi: E \rightarrow C$ such that $\varphi|_K = \sigma$.*

PROOF. Suppose first that $E = K(x)$ and let $f = f(x, K)$. Let $\bar{\sigma}(f) \in C[X]$ and let $y \in C$ be a root of $\bar{\sigma}(f)$. If $z \in E$, then $z = g(x)$ for some $g \in K[X]$. Let $\varphi: E \rightarrow C$, $z \mapsto \bar{\sigma}(g)(y)$.

The map φ is well-defined. If $z = h(x)$ for some $h \in K[X]$, then

$$0 = g(x) - h(x) = (g - h)(x)$$

and thus f divides $g - h$. In particular, $\bar{\sigma}(f)$ divides $\bar{\sigma}(g - h) = \bar{\sigma}(g) - \bar{\sigma}(h)$ and hence

$$(\bar{\sigma}(g) - \bar{\sigma}(h))(y) = 0.$$

It is an exercise to show that the map φ is a ring homomorphism.

Let $a \in K$. It follows that $\varphi|_K = \sigma$, as

$$\varphi(a) = \bar{\sigma}(aX^0)(y) = \sigma(a)$$

Let us now prove the proposition in full generality. Let X be the set of pairs (F, τ) , where F is a subfield of E that contains K and $\tau: F \rightarrow C$ is a field homomorphism such that $\tau|_K = \sigma$. Note that $(K, \sigma) \in X$, so X is non-empty. Moreover, X is partially ordered by

$$(F, \tau) \leq (F_1, \tau_1) \iff F \subseteq F_1 \text{ and } \tau_1|_F = \tau.$$

If $\{(F_i, \tau_i) : i \in I\}$ is a chain in X , then $F = \cup_{i \in I} F_i$ is a subfield of E that contains K . Moreover, if $z \in F$, then $z \in F_i$ for some $i \in I$ and then one defines $\tau(z) = \tau_i(z)$. It is an exercise to prove that τ is well-defined. Since $(F, \tau) \in X$ is an upper bound, Zorn's lemma implies that there exists a maximal element $(E_1, \theta) \in X$. We claim that $E = E_1$. If not, let $z \in E \setminus E_1$. Since we know the proposition is true for the extension $E_1(z)/E_1$, let $\rho: E_1(z) \rightarrow C$ be a field homomorphism such that $\rho|_{E_1} = \theta$. Then, in particular, $\rho|_K = \sigma$. This implies that $(E_1(z), \rho) \in X$ and hence $(E_1, \theta) < (E_1(z), \rho)$, a contradiction to the maximality of (E_1, θ) . $\square$

The previous proposition will be used to prove that the algebraic closure is unique up to isomorphism.

3.3. THEOREM (Artin). *Let K be a field. Then K admits an algebraic closure C/K . If C_1/K is an algebraic closure, then the extensions C/K and C_1/K are isomorphic.*

§ 3. Artin's theorem

PROOF. Let us first prove the uniqueness. The previous proposition implies the existence of an extension homomorphism $\varphi: C \rightarrow C_1$. Let $y \in C_1$ and $f = f(y, K)$ be the minimal polynomial of y in K . Since f admits a factorization

$$f = \prod (X - \alpha_i)^{m_i}$$

in $C[X]$, it follows that

$$f = \overline{\varphi}(f) = \prod (X - \varphi(\alpha_i))^{m_i}$$

Since $0 = f(y)$, we conclude that $y = \varphi(\alpha_j)$ for some j . In particular, φ is surjective and hence φ is bijective.

We now prove the existence. Let us assume that K admits an extension E/K with E algebraically closed. We will prove later that this extension indeed exists; at the moment, we only want to get an algebraic extension from this setting. Let

$$F = \{x \in E : x \text{ is algebraic over } K\}.$$

Then F/K is algebraic. Let $g \in F[X]$ be such that $\deg g > 0$. Since E is algebraically closed, g admits a root α in E . In particular, α is algebraic over F and hence α is algebraic over K . This implies that $\alpha \in F$, thus F is algebraically closed. This proves that F/K is an algebraic closure.

Let us prove that there exists an extension E_1/K such that every polynomial $f \in K[X]$ with $\deg f > 0$ has a root in E_1 . Let $\{f_i : i \in I\}$ be the family of monic irreducible polynomials with coefficients in K . We may think that $f_i = f_i(X_i)$. Let $R = K[\{X_i : i \in I\}]$ and let J be the ideal of R generated by the $f_i(X_i)$. We claim that $J \neq R$. If not, $1 \in J$, so

$$1 = \sum_{j=1}^m g_j f_{i_j}(X_{i_j})$$

for some $g_1, \dots, g_m \in R$. There exists an extension F/K such that f_{i_j} has a root α_j in F for all j . Let

$$\tau: R \rightarrow F, \quad \tau(X_k) = \begin{cases} \alpha_j & \text{if } k = i_j, \\ 0 & \text{if } k \notin \{i_1, \dots, i_m\}. \end{cases}$$

Then τ is a ring homomorphism and

$$1 = \tau(1) = \sum_{j=1}^m \tau(g_j) f_{i_j}(\alpha_j) = 0,$$

a contradiction.

Since J is a proper ideal, it is contained in a maximal ideal M . Let $L = R/M$ and let $\sigma: K \rightarrow L$ be the composition $K \hookrightarrow R \rightarrow R/M = L$, where $\pi: R \rightarrow R/M$ is the canonical map. As we did before, $\pi(X_i)$ is a root of $\overline{\sigma}(f_i)$ for all i . And there exists an extension E_1/K such that every f_i has a root in E_1 . Proceeding in this way, we construct a sequence

$$E_1 \subseteq E_2 \subseteq \dots$$

of fields such that every polynomial of positive degree and coefficients in E_k admits a root in E_{k+1} . Let $E = \cup E_k$. We claim that E is algebraically closed. In fact, let $g \in E[X]$ be such that $\deg g > 0$. Then, since $g \in E_r[X]$ for some r , it follows that g has a root in $E_{r+1} \subseteq E$. $\square$

§ 4. Splitting fields

4.1. DEFINITION. Let K be a field and $f \in K[X]$ be such that $\deg f > 0$. A **splitting field** of f over K is a field E that contains K and that satisfies the following properties:

- 1) f factorizes linearly in $E[X]$.
- 2) If F is a field such that $K \subseteq F \subseteq E$ and f factorizes linearly in $F[X]$, then $F = E$.

Easy examples:

4.2. EXAMPLE. $\mathbb{C}$ is a splitting field of $X^2 + 1 \in \mathbb{R}[X]$.

4.3. EXAMPLE. $\mathbb{Q}[\sqrt{2}]$ is a splitting field of $X^2 - 2 \in \mathbb{Q}[X]$.

4.4. EXAMPLE. The splitting field of $f = X^2 - 2$ over $\mathbb{Z}/7$ is precisely $\mathbb{Z}/7$, as 3 and 4 are the roots of f in $\mathbb{Z}/7$.

4.5. EXAMPLE. $\mathbb{Q}(\sqrt[3]{2})$ is not a splitting field of $X^3 - 2 \in \mathbb{Q}[X]$. However, if ω is a primitive cubic root of one, then $\mathbb{Q}(\sqrt[3]{2}, \omega)$ is a splitting field of the polynomial $X^3 - 2 \in \mathbb{Q}[X]$.

4.6. PROPOSITION. *E is a splitting field of $f \in K[X]$ if and only if f factorizes linearly in $E[X]$ and $E = K(x_1, \dots, x_n)$, where $x_1, \dots, x_n$ are the roots of f .*

PROOF. Let $f = a \prod_{i=1}^r (X - x_i)^{n_i}$ and $F = K(x_1, \dots, x_r)$ with $x_1, \dots, x_r \in E$. Since f factorizes linearly in $F[X]$, it follows that $F = E$. Conversely, let $E = K(x_1, \dots, x_r)$ and assume that f factorizes linearly in $F[X]$. Then, in particular, $x_1, \dots, x_r \in F$. Hence $E \subseteq F$ and $F = E$. $\square$

Note that if $E = K(x_1, \dots, x_n)$ as in the statement of Proposition 4.6, then it automatically follows that f factorizes linearly in $E[X]$.

One immediately obtains the following consequence: If E is a splitting field of $f \in K[X]$, then E/K is finite.

4.7. THEOREM. *Let $f \in K[X]$ be such that $\deg f > 0$. There exists a (unique up to extension isomorphism) splitting field of f over K .*

PROOF. Let C/K be an algebraic closure of K . Write

$$f = a \prod_{i=1}^r (X - x_i)^{n_i}$$

in $C[X]$. Then $E = K(x_1, \dots, x_r)$ is a splitting field of f over K .

Let us prove the uniqueness: if E_1/K is a splitting field of f over K , then E_1/K is algebraic and thus Proposition 3.2 implies that there exists $\varphi \in \text{Hom}(E_1/K, C/K)$, that is $\varphi: E_1 \rightarrow C$ is a field homomorphism such that $\varphi|_K$ is the identity. Factorize f linearly in $E_1[X]$ and apply φ :

$$f = a \prod_{j=1}^s (X - y_j)^{m_j} \implies f = \varphi(f) = \varphi(a) \prod_{j=1}^s (X - \varphi(y_j))^{m_j}$$

so f factorizes linearly in $\varphi(E_1)[X]$. Moreover, $E_1 = K(y_1, \dots, y_s)$ and

$$\varphi(E_1) = K(\varphi(y_1), \dots, \varphi(y_s)).$$

Thus $\varphi(E_1)$ is a splitting field of f . Since $\varphi(E_1) \subseteq E$, it follows that $\varphi(E_1) = E$. $\square$

§ 4. Splitting fields

4.8. EXERCISE. If C is an algebraic closure of K and $\varphi \in \text{Hom}(C/K, C/K)$, then φ is an isomorphism.

Let C be an algebraic closure of K and $G = \text{Gal}(C/K)$. The group G acts on C

$$\sigma \cdot x = \sigma(x), \quad \sigma \in G, x \in C.$$

The orbits are of the form

$$O_G(x) = \{\sigma(x) : \sigma \in G\} = \{y \in C : y = \sigma(x) \text{ for some } \sigma \in G\}$$

The elements $x, y \in C$ are **conjugate** if $y = \sigma(x)$ for some $\sigma \in G$.

4.9. PROPOSITION. *Let C be an algebraic closure of K and $x, y \in C$. Then x and y are conjugate if and only if $f(x, K) = f(y, K)$. In particular, $O_G(x)$ is finite.*

PROOF. Let $G = \text{Gal}(C/K)$. If x and y are conjugate, say $y = \sigma(x)$ for some $\sigma \in G$, let us write $g = f(x, K)$ as

$$g = \sum_{i=0}^n a_i X^i$$

for some $n \geq 1$ and $a_0, \dots, a_n \in K$. Then $0 = g(x) = \sum_{i=0}^n a_i x^i$ and hence y is a root of g , as

$$\begin{aligned} 0 &= \sigma \left(\sum_{i=0}^n a_i x^i \right) = \sum_{i=0}^n \sigma(a_i) \sigma(x)^i \\ &= \sum_{i=0}^n a_i \sigma(x)^i = \sum_{i=0}^n a_i y^i. \end{aligned}$$

Thus $f(y, K) = g$.

Conversely, assume that $f(x, K) = f(y, K)$. Let $g = f(x, K) = f(y, K)$ and let

$$\varphi: K[x] \rightarrow K[y], \quad h(x) \mapsto h(y).$$

Let us show that the map φ is well-defined: we need to show that if $h_1(x) = h_2(x)$, then

$$h_1(y) = \varphi(h_1(x)) = \varphi(h_2(x)) = h_2(y).$$

If $h_1(x) = h_2(x)$, then

$$(h_1 - h_2)(x) = h_1(x) - h_2(x) = 0.$$

This implies that g divides $h_1 - h_2$. In particular, $h_1(y) = h_2(y)$.

A straightforward calculation shows that φ is a field homomorphism such that $\varphi|_K = \text{id}$, this means that φ is an extension homomorphism such that $\varphi(x) = y$. There exists a homomorphism $\sigma \in \text{Hom}(C/K, C/K)$ such that $\sigma|_{K[x]} = \varphi$. Since σ is bijective (why?), $\sigma(x) = \varphi(x) = y$ and hence $O_G(x) = O_G(y)$. $\square$

4.10. PROPOSITION. *Let C be an algebraic closure of K and $x \in C$. Then*

$$f(x, K) = \prod_{y \in O_G(x)} (X - y)^m$$

for some m .

§ 5. Normal extensions

PROOF. For each $y \in O_G(x)$ let m_y be the multiplicity of y in $f(x, K)$. Then, for example, $f(x, K) = (X - x)^{m_x}g$ for some g . If $y \in O_G(x)$, then $y = \sigma(x)$ for some $\sigma \in \text{Gal}(C/K)$. Since

$$\bar{\sigma}(f(x, K)) = f(x, K) = (X - y)^{m_x}\bar{\sigma}(g),$$

it follows that $m_y \geq m_x$. By symmetry, we conclude that $m_x = m_y$. $\square$

The previous proposition shows, in particular, that all the roots of an irreducible polynomial $f \in K[X]$ in an algebraic closure C of K have the same multiplicity. This is not true if f is not irreducible. Find an example.

4.11. DEFINITION. Let K be a field and $\{f_i : i \in I\}$ be a non-empty family of polynomials of positive degree with coefficients in K . A **splitting field** of $\{f_i : i \in I\}$ is an extension E/K such that every f_i factorizes linearly in $E[X]$ and if F/K is a subextension of E/K such that every f_i factorizes linearly in $F[X]$, then $F = E$.

4.12. EXERCISE. Prove that E/K is a splitting field of $\{f_i : i \in I\}$ if and only if every f_i factorizes linearly in $E[X]$ and $E = K(S)$ where $S = \{\text{roots of } f_i \text{ for all } i\}$.

4.13. EXERCISE. Prove that if E/K is a splitting field of $\{f_i : i \in I\}$, then E/K is algebraic. If, moreover, I is finite, then E/K is a splitting field of $\prod_{i \in I} f_i$.

4.14. EXERCISE. Prove that there exists a splitting field of $\{f_i : i \in I\}$ and it is unique up to extension isomorphism.

4.15. EXERCISE. Let $f = X^3 - X - 1 \in (\mathbb{Z}/3)[X]$ and E be a splitting field of f . Compute $[E : \mathbb{Z}/3]$.

What about the splitting field of $f = X^3 - X - 1 \in \mathbb{Q}[X]$?

4.16. EXERCISE. Let $f = X^4 - 5X^2 + 5 \in \mathbb{Q}[X]$ and E be a splitting field of f . Compute $[E : \mathbb{Q}]$ and $\text{Gal}(E/\mathbb{Q})$.

§ 5. Normal extensions

5.1. PROPOSITION. Let E/K be an algebraic extension and $\sigma \in \text{Hom}(E/K, E/K)$. Then σ is bijective.

PROOF. It is enough to prove that σ is surjective. Why?

Let $x \in E$, $f = f(x, K)$, and let $S = \{y \in E : f(y) = 0\}$. Note that for any $y \in S$, we have that $\sigma(y) \in S$, and hence $\sigma(S) \subseteq S$. However, as σ is injective and S is finite, we must have that $\sigma(S) = S$. Hence there exists a $y \in S \subseteq E$ such that $\sigma(y) = x$, and therefore σ must be surjective. $\square$

5.2. DEFINITION. Let E/K be an algebraic extension and C be an algebraic closure of K containing E . Then E/K is **normal** if $\sigma(E) \subseteq E$ for all $\sigma \in \text{Hom}(E/K, C/K)$.

Note that $\sigma(E) \subseteq E$ in the previous definition is equivalent to $\sigma(E) = E$.

§ 5. Normal extensions

5.3. EXAMPLE. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal. Why?

Some trivial examples of normal extensions: K/K is normal and if C is an algebraic closure of K , then C/K is normal.

5.4. EXAMPLE. The extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is normal. Every extension generated by algebraic elements of degree two is normal.

5.5. EXERCISE. Let ξ be a primitive cubic root of one. Then $\mathbb{Q}(\sqrt[3]{2}, \xi)/\mathbb{Q}$ is normal.

The following result is practical but technical. That is why we leave the proof as an exercise.

5.6. EXERCISE. Prove that the previous definition depends only on E (and not on the algebraic closure C).

Some properties:

5.7. PROPOSITION. Let E/K be a normal extension and $f \in K[X]$ be an irreducible polynomial that admits a root x in E . Then f factorizes linearly in E .

PROOF. We may assume that f is monic. Let C/K be an algebraic closure of K containing E . Let y be a root of f in C . Since $f = f(x, K) = f(y, K)$, it follows that $y = \sigma(x)$ for some $\sigma \in \text{Gal}(C/K)$. Since E/K is normal, $\sigma|_E: E \rightarrow C$ is an automorphism of E/K , that is $\sigma(E) \subseteq E$. In particular, $y \in E$. $\square$

Let $K \subseteq F \subseteq E$ be a tower of fields. If E/K is normal, then E/F is normal. However, Note that E/K normal does not imply F/K normal, as this would imply that every extension is normal. Moreover, E/F normal and F/K normal do not imply E/K normal.

5.8. EXAMPLE. The extensions $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ are both normal, but the extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not, as the roots of $X^4 - 2$ are $\sqrt[4]{2}$, $-\sqrt[4]{2}$, $\sqrt[4]{2}i$ and $-\sqrt[4]{2}i$.

Recall that if C is an algebraic closure of K and $x \in C$, then

$$f(x, K) = \prod (X - y)^m,$$

where the product is taken over all $y \in O_{\text{Gal}(C/K)}(x)$. If E/K is normal and $x \in E$, then there exists m such that

$$f(x, K) = \prod (X - y)^m,$$

where the product is taken over all $y \in O_{\text{Gal}(E/K)}(x)$.

5.9. PROPOSITION. Let E/K and F/K be extensions. If F/K is normal, then EF/E is normal.

PROOF. Let C be an algebraic closure of E containing EF (this exists because EF/E is algebraic). Let $\sigma \in \text{Hom}(EF/E, C/E)$. We claim that $\sigma(EF) = EF$. Let

$$\overline{K} = \{x \in C : x \text{ is algebraic over } K\}.$$

Then $\overline{K}$ is an algebraic closure over K and $F \subseteq \overline{K}$. Since F/K is normal and $\sigma|_F \in \text{Hom}(F/K, \overline{K}/K)$, it follows that $\sigma(F) = F$. If $z \in EF$, then $z = \sum_{i=1}^m e_i f_i$ for some

§ 6. Dedekind's theorem

$e_1, \dots, e_m \in E$ and $f_1, \dots, f_m \in F$. Since $\sigma(e_i) = e_i$ for all i ,

$$\sigma(z) = \sum_{i=1}^m \sigma(e_i) \sigma(f_i) = \sum_{i=1}^m e_i \sigma(f_i) \in EF. \quad \square$$

What is the relation between normal extensions and splitting fields? The notions look deeply related. The following proposition serves as an explanation:

5.10. PROPOSITION. *Let E/K be an algebraic extension. Then E/K is normal if and only if E/K is the splitting field of a family of polynomials of $K[X]$ of positive degree.*

PROOF. Assume first that E/K is a normal extension. Let $G = \text{Gal}(E/K)$. If $x \in E$ and $f(x, K) = \prod_{y \in O_G(x)} (X - y)^m$, then $f(x, K)$ factorizes linearly in $E[X]$. Thus E/K is a splitting field of the family $\{f(x, K) : x \in E\}$.

Conversely, assume that E/K is a splitting field of the family $\{f_i : i \in I\}$. Then $E = K(S)$ where S is the set of roots of the polynomials f_i . Let C/K be an algebraic closure of K that contains E and let $\sigma \in \text{Hom}(E/K, C/K)$. Let $x \in S$. Then x is a root of some $f_j = \sum a_k X^k$. Since $f_j(x) = 0$, it follows that $\sigma(x)$ is a root of f_j , as

$$f_j(\sigma(x)) = \sum a_k \sigma(x)^k = \sum \sigma(a_k) \sigma(x^k) = \sigma\left(\sum a_k x^k\right) = \sigma(0) = 0.$$

Hence $\sigma(E) \subseteq E$. $\square$

5.11. EXERCISE. Let $E = \mathbb{Q}[\sqrt[4]{7} + \sqrt{2}]$.

- 1) Prove that $E/\mathbb{Q}$ is not normal.
- 2) Compute $[E : \mathbb{Q}]$.
- 3) Compute $\text{Gal}(E/\mathbb{Q})$.

§ 6. Dedekind's theorem

Note that every extension homomorphism $E/K \rightarrow F/K$ is, in particular, a K -linear map $E \rightarrow F$, that is

$$\text{Hom}(E/K, F/K) \subseteq \text{Hom}_K(E, F).$$

If F/K is an extension and V is a K -vector space, the set $\text{Hom}_K(V, F)$ of K -linear maps is a vector space over F with $(a \cdot f)(v) = af(v)$ for $a \in F$, $f \in \text{Hom}_K(V, F)$ and $v \in V$.

6.1. EXERCISE. Let V be a K -vector space. Prove that $\dim_F \text{Hom}_K(V, F) \geq \dim_K V$. Moreover, if $\dim_K V < \infty$, then $\dim_F \text{Hom}_K(V, F) = \dim_K V$.

If V is a vector space and S is a (possibly infinite) subset of V , then S is linearly independent if every finite subset of S is linearly independent.

6.2. THEOREM (Dedekind). *Let E/K and F/K be extensions and let $\{\varphi_i : i \in I\}$ be a subset of $\text{Hom}(E/K, F/K)$, i.e. a family of extension homomorphisms. Assume that $\varphi_i \neq \varphi_j$ if $i \neq j$. Then the subset $\{\varphi_i : i \in I\} \subseteq \text{Hom}_K(E, F)$ is linearly independent over F .*

PROOF. Assume it is not. Let $\{\varphi_1, \dots, \varphi_n\}$ be linearly dependent over F with n minimal. Clearly, $n > 1$. Without loss of generality, we may assume that

$$(6.1) \quad \sum_{i=1}^n a_i \varphi_i = 0$$

§ 6. Dedekind's theorem

for some $a_1, \dots, a_n \in F$ all different from zero. Let $z \in E \setminus \{0\}$ be such that $\varphi_1(z) \neq \varphi_2(z)$. If $x \in E$, then

$$0 = \left(\sum_{i=1}^n a_i \varphi_i \right) (xz) = \sum_{i=1}^n a_i \varphi_i(xz) = \sum_{i=1}^n a_i \varphi_i(x) \varphi_i(z) = \left(\sum_{i=1}^n (a_i \varphi_i(z)) \varphi_i \right) (x).$$

Thus

$$\sum_{i=1}^n (a_i \varphi_i(z)) \varphi_i = 0.$$

Since $\varphi_1(z) \neq 0$,

$$(6.2) \quad a_1 \varphi_1 + a_2 \frac{\varphi_2(z)}{\varphi_1(z)} \varphi_2 + \dots + a_n \frac{\varphi_n(z)}{\varphi_1(z)} \varphi_n = 0.$$

Thus, subtracting (6.1) and (6.2),

$$\left(a_2 - a_2 \frac{\varphi_2(z)}{\varphi_1(z)} \right) \varphi_2 + \dots + \left(a_n - a_n \frac{\varphi_n(z)}{\varphi_1(z)} \right) \varphi_n = 0.$$

Since $a_2 \neq 0$ and $\varphi_2(z) \neq \varphi_1(z)$, the scalar $a_2 - a_2 \frac{\varphi_2(z)}{\varphi_1(z)} \neq 0$ and hence $\{\varphi_2, \dots, \varphi_n\}$ is linearly dependent, a contradiction. $\square$

If E/K and F/K are extensions, let $\gamma(E/K, F/K) = |\text{Hom}(E/K, F/K)|$.

6.3. EXERCISE. Prove the following statements:

- 1) $\gamma(E/K, F/K) \leq \dim_F \text{Hom}_K(E, F)$.
- 2) If $[E : K] < \infty$, then $\gamma(E/K, F/K) \leq [E : K]$.
- 3) If x is algebraic over K , then $\gamma(K(x)/K, F/K) \leq \deg f(x, K)$.

If C is an algebraic closure of K , then we define $\gamma(E/K) = \gamma(E/K, C/K)$. This definition does not depend on the algebraic closure.

6.4. EXERCISE. If C and C_1 are algebraic closures of K , then

$$|\text{Hom}(E/K, C/K)| = |\text{Hom}(E/K, C_1/K)|.$$

6.5. PROPOSITION. Let C be an algebraic closure of K and $G = \text{Gal}(C/K)$. If $x \in C$, then $\gamma(K(x)/K) = |O_G(x)|$.

PROOF. If $\sigma \in \text{Hom}(K(x)/K, C/K)$, then there exists $\phi \in G$ such that $\phi|_{K(x)} = \sigma$. Thus

$$\sigma(x) = \phi(x) \in O_G(x).$$

Conversely, if $y \in O_G(x)$, then there exists $\tau \in G$ such that $y = \tau(x)$. Hence

$$\tau|_{K(x)} \in \text{Hom}(K(x)/K, C/K)$$

and $\tau|_{K(x)}(x) = y$. Since our sets are then in bijective correspondence, the claim follows. $\square$

6.6. EXERCISE. If E/K is finite, then $|\text{Gal}(E/K)| \leq [E : K]$. Moreover, E/K is normal if and only if $|\text{Gal}(E/K)| = \gamma(E/K)$.

§ 6. Dedekind's theorem

If $t: A \rightarrow B$ is a surjective map, then $a \sim a_1 \iff t(a) = t(a_1)$ defines an equivalence relation on A . The set $\bar{A}$ of equivalence classes is in bijective correspondence with B , $\bar{A} \rightarrow B$, $\bar{a} \mapsto t(a)$. Moreover, if $|t^{-1}(\{b\})| = m$ for all $b \in B$, then $|A| = m|\bar{A}| = m|B|$.

6.7. PROPOSITION. *Let E/K be algebraic and F/K be a subextension such that E/F is finite. Then $\gamma(E/K) = \gamma(E/F)\gamma(F/K)$.*

PROOF. Assume first that $E = F(x)$. Let C be an algebraic closure of K containing E and $G = \text{Gal}(C/F)$. Let $f = f(x, F) = \sum b_i X^i$.

The map

$$\lambda: \text{Hom}(E/K, C/K) \rightarrow \text{Hom}(F/K, C/K), \quad \sigma \mapsto \sigma|_F,$$

is well-defined. It is surjective: if $\varphi \in \text{Hom}(F/K, C/K)$, then $\varphi: F \rightarrow C$ is, in particular, a field homomorphism. Since E/F is algebraic, by Proposition 3.2 there exists a field homomorphism $\sigma: E \rightarrow C$ such that $\sigma|_F = \varphi$. Since $\sigma|_K = \varphi|_K = \text{id}$, in particular $\sigma \in \text{Hom}(E/K, C/K)$.

For $\varphi \in \text{Hom}(F/K, C/K)$,

$$\lambda^{-1}(\{\varphi\}) = \{\sigma \in \text{Hom}(E/K, C/K) : \sigma|_F = \varphi\}$$

and let R_φ be the set of roots (in C) of the polynomial $\bar{\varphi}(f) = \sum \varphi(b_i)X^i$.

CLAIM. The map $\alpha: \lambda^{-1}(\{\varphi\}) \rightarrow R_\varphi$, $\sigma \mapsto \sigma(x)$, is well-defined.

We need to show that $\sigma(x)$ is a root of $\bar{\varphi}(f)$:

$$\begin{aligned} \bar{\varphi}(f)(\sigma(x)) &= \sum \varphi(b_i)\sigma(x)^i = \sum \sigma(b_i)\sigma(x)^i \\ &= \sum \sigma(b_i x^i) = \sigma\left(\sum b_i x^i\right) = \sigma(f(x)) = \sigma(0) = 0. \end{aligned}$$

CLAIM. The map $\beta: R_\varphi \rightarrow \lambda^{-1}(\{\varphi\})$, $y \mapsto \sigma_y$, where $\sigma_y(z) = \bar{\varphi}(h)(y)$ if $z = h(x)$, is well-defined.

We need to show that if $z = h(x)$ and $z = h_1(x)$ for some $h, h_1 \in F[X]$, then

$$\bar{\varphi}(h)(y) = \bar{\varphi}(h_1)(y).$$

The assumptions imply that $(h - h_1)(x) = 0$ and hence f divides $h - h_1$. Since $\bar{\varphi}$ is a ring homomorphism, $\bar{\varphi}(f)$ divides $\bar{\varphi}(h) - \bar{\varphi}(h_1)$. This implies $(\bar{\varphi}(h) - \bar{\varphi}(h_1))(y) = 0$. We also need to show that $\sigma_y|_F = \varphi$: if $a \in F$, then write $a = aX^0 \in F[X]$. Thus $\sigma_y(a) = \bar{\varphi}(aX^0)(y) = \varphi(a) \in C$. It is now an exercise to prove that $\sigma_y \in \text{Hom}(E/K, C/K)$.

CLAIM. $|\lambda^{-1}(\{\varphi\})| = |R_\varphi|$.

For this we need to show that β is the inverse of α , that is $\alpha \circ \beta = \text{id}$ and $\beta \circ \alpha = \text{id}$. To prove that $\beta \circ \alpha = \text{id}$ let σ be such that $\sigma|_F = \varphi$. Then $y = \sigma(x) \in R_\varphi$. Let

$$z = h(x) = \sum a_i x^i \in F[x] = E.$$

Then

$$\bar{\varphi}(h)(y) = \sum \varphi(a_i)y^i = \sum \sigma(a_i)y^i = \sigma\left(\sum a_i x^i\right) = \sigma(z).$$

Conversely, if $y \in R_\varphi$, then

$$\alpha(\sigma_y) = \sigma_y(x) = y,$$

as $\sigma_y(x) = \bar{\varphi}(X)(y) = y$.

§ 7. Separable extensions

CLAIM. If $\phi \in \text{Gal}(C/K)$ is such that $\phi|_F = \varphi$, then $|\phi^{-1}(R_\varphi)| = |R_\varphi|$ and

$$O_G(x) = \phi^{-1}(R_\varphi).$$

Let us first prove $O_G(x) \supseteq \phi^{-1}(R_\varphi)$. If $y \in R_\varphi$, then

$$\begin{aligned} f(\phi^{-1}(y)) &= \sum b_i \phi^{-1}(y^i) = \phi^{-1} \left(\sum \phi(b_i) y^i \right) \\ &= \phi^{-1}(\overline{\varphi}(f)(y)) = \phi^{-1}(0) = 0. \end{aligned}$$

Then $f(x, F) = f(\phi^{-1}(y), F)$. By Proposition 4.9, $\phi^{-1}(y) \in O_G(x)$.

Now we prove $O_G(x) \subseteq \phi^{-1}(R_\varphi)$. Let $z \in O_G(x)$. Then $\overline{\varphi}(f)(\phi(z)) = 0$, as

$$\begin{aligned} \overline{\varphi}(f)(\phi(z)) &= \sum \varphi(b_i) \phi(z^i) \\ &= \sum \phi(b_i) \phi(z^i) = \phi \left(\sum b_i z^i \right) = \phi(f(z)) = \phi(0) = 0. \end{aligned}$$

Thus $\phi(z) \in R_\varphi$ and hence $z \in \phi^{-1}(R_\varphi)$. It follows that $|\lambda^{-1}(\{\varphi\})| = |O_G(x)|$ for all φ . By using the argument before the proposition,

$$\begin{aligned} \gamma(E/K) &= |\text{Hom}(E/K, C/K)| \\ &= |O_G(x)| |\text{Hom}(F/K, C/K)| \\ &= |O_G(x)| \gamma(F/K). \end{aligned}$$

Since $\gamma(E/F) = \gamma(F(x)/F) = |O_G(x)|$ by Proposition 6.5, the claim follows.

For the general case, we assume that $E = F(x_1, \dots, x_n)$. We proceed by induction on n . If $n = 0$, then $E = F$ and the result is trivial. If $n > 0$, let $L = F[x_1, \dots, x_{n-1}]$ and $E = L(x_n)$. The case proved implies that $\gamma(E/F) = \gamma(E/L)\gamma(L/F)$. By the inductive hypothesis, $\gamma(L/K) = \gamma(L/F)\gamma(F/K)$. Thus

$$\gamma(E/F)\gamma(F/K) = \gamma(E/L)\gamma(L/F)\gamma(F/K) = \gamma(E/L)\gamma(L/K) = \gamma(E/K),$$

again using the previous case. □

§ 7. Separable extensions

7.1. DEFINITION. Let E/K be an extension and $x \in E$ an algebraic element over K . Then x is **separable** over K if x is a simple root of $f(x, K)$.

An algebraic extension E/K is **separable** if every $x \in E$ is separable over K . Then K/K is separable.

7.2. EXERCISE. Prove that an element x is separable over K if and only if x is a simple root of a polynomial with coefficients in K .

If F/K is a subextension of E/K and $x \in E$ is separable over K , then x is separable over F .

§ 7. Separable extensions

7.3. EXERCISE. If C is an algebraic closure of K , $x \in C$ and $G = \text{Gal}(C/K)$. Prove that the following statements are equivalent:

- 1) x is separable over K .
- 2) Every $y \in O_G(x)$ is separable over K .
- 3) $\gamma(K(x)/K) = [K(x) : K] = \deg f(x, K)$.

Let K be any field and $g \in K[X]$. Let z be a root of g . Then z is a multiple root of g if and only if z is a root of g' .

7.4. EXERCISE. Prove that if K has characteristic zero or K is finite, then every algebraic extension of K is separable.

7.5. EXAMPLE. Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Then $[E : \mathbb{Q}] = 4$ and $\text{Gal}(E/\mathbb{Q}) \simeq C_2 \times C_2$. The extension $E/\mathbb{Q}$ is normal, as it is the splitting field of $(X^2 - 2)(X^2 - 3)$ and it is separable as $\mathbb{Q}$ has characteristic zero.

7.6. EXAMPLE. Let E be a splitting field of $X^4 - 2$ over $\mathbb{Q}$. Then $E/\mathbb{Q}$ is normal and separable. Note that $E = \mathbb{Q}(\sqrt[4]{2}, i)$, so

$$[E : \mathbb{Q}] = 8 = |\text{Gal}(E/\mathbb{Q})|.$$

Let us compute $\text{Gal}(E/\mathbb{Q})$. If $\sigma \in \text{Gal}(E/\mathbb{Q})$, then $\sigma(\sqrt[4]{2}) \in \{\sqrt[4]{2}, -\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}i\}$ and $\sigma(i) \in \{-i, i\}$. Two examples are

$$\alpha: \begin{cases} \sqrt[4]{2} \mapsto \sqrt[4]{2}i, \\ i \mapsto i, \end{cases} \quad \beta: \begin{cases} \sqrt[4]{2} \mapsto \sqrt[4]{2}, \\ i \mapsto -i. \end{cases}$$

It follows that $\text{Gal}(E/\mathbb{Q})$ is isomorphic to the group $\langle \alpha, \beta \rangle$, which turns out to be isomorphic to the dihedral group of eight elements.

Another consequence: If $E = K(S)$, then E/K is separable if and only if every $x \in S$ is separable over K . One first does the case $E = K(x)$ and then proceeds by induction.

7.7. EXERCISE. Let $K \subseteq F \subseteq E$ be a tower of fields. Prove that E/K is separable if and only if F/K and E/F are separable.

7.8. EXERCISE. Let E/K and F/K be extensions. Prove that if F/K is separable, then EF/E is separable.

7.9. EXAMPLE. Let p be a prime and let t be transcendental over $\mathbb{F}_p = \mathbb{Z}/p$. Let $K = \mathbb{F}_p(t^p)$ and let $E = \mathbb{F}_p(t)$. Then E/K has degree p and is not separable.

Indeed, E/K is algebraic because $f = f(t, K) = X^p - t^p$ (it is an exercise to check that f is irreducible in $K[X]$). Moreover, $f = (X - t)^p \in E[X]$, and so t has multiplicity p in f . It follows that t is not separable over K . Otherwise there is a polynomial $g \in K[X]$ such that t is a simple root of g , and hence, in particular, $g(t) = 0$. However, we then must have $f \mid g$, and thus t cannot be a simple root of g .

Note that, in this case, E/K is normal, as E is the splitting field of f over K .

If E/K is algebraic, then

$$F = \{x \in E : x \text{ is separable over } K\}$$

§ 8. Steinitz' theorem

is a subfield of E that contains K . It is known as the **separable closure** of K with respect to E . Note that $F = K(F)$, as $K(F)$ is separable because it is generated by separable elements. Moreover, F/K is separable and E/F is a **purely inseparable** extension, meaning that for every $x \in E \setminus F$, x is not separable over K .

7.10. PROPOSITION. *If E/K is separable and finite, then $E = K(x)$ for some $x \in E$.*

PROOF. Let us assume that K is finite. Then E is finite and hence the multiplicative group $E^\times = E \setminus \{0\}$ is cyclic, say $E^\times = \langle x \rangle$. It follows that $E = K(x)$.

Let us now assume that K is infinite. We first consider the case $E = K(x, y)$. The general case $E = K(x_1, \dots, x_n)$ is left as an exercise, one needs to proceed by induction. Let $n = [E : K]$ and C be an algebraic closure of K containing E . Write $\text{Hom}(E/K, C/K) = \{\sigma_1, \dots, \sigma_n\}$. Let

$$f = \prod_{1 \leq i < j \leq n} ((\sigma_i(y) - \sigma_j(y)) + X(\sigma_i(x) - \sigma_j(x))) \in C[X].$$

Then $f \neq 0$, as f is a product of non-zero polynomials. Since K is infinite, there exists a non-zero $c \in K$ such that $f(c) \neq 0$. For any $r, s \in \{1, \dots, n\}$ with $r \neq s$,

$$\sigma_r(y) - \sigma_s(y) + c(\sigma_r(x) - \sigma_s(x)) \neq 0,$$

as $f(c) \neq 0$. It follows that $\sigma_r(y + cx) \neq \sigma_s(y + cx)$. Thus $\gamma(K(y + cx)/K) \geq n$. Now

$$n \geq [K(y + cx) : K] = \gamma(K(y + cx)/K) \geq n,$$

so $[K(y + cx) : K] = n$ and hence $K(y + cx) = E$. □

For example, $\mathbb{Q}(\sqrt{2}, i) = \mathbb{Q}(\sqrt{2} + i)$.

§ 8. Steinitz' theorem

8.1. THEOREM (Steinitz). *Let E/K be a finite extension. Then $E = K(x)$ for some $x \in E$ if and only if E/K admits finitely many subextensions.*

PROOF. We may assume that K is infinite; otherwise, the result is trivial. We first prove $\implies$. Let us assume that $E = K(x)$ for some x . We claim that the map

$$\begin{aligned} \Psi: \{F : K \subseteq F \subseteq E\} &\rightarrow \{g \in E[X] : g \text{ is a monic divisor of } f(x, K)\}, \\ F &\mapsto f(x, F), \end{aligned}$$

is injective. Take F_0 such that $K \subseteq F_0 \subseteq F \subseteq E$ and $f(x, F) = f(x, F_0)$. Then

$$[E : F_0] = [F_0(x) : F_0] = \deg f(x, F_0) = m = [F(x) : F] = [E : F]$$

and hence $F = F_0$.

In general, let F_1 and F_2 be such that $K \subseteq F_1, F_2 \subseteq E$ and $f(x, F_1) = f(x, F_2)$. Let $F_0 = F_1 \cap F_2$. Then $f = f(x, F_1) = f(x, F_2) \in F_0[X]$ and hence $f(x, F_0) = f$. Hence we can apply what we proved before to $F_0 \subseteq F_1$ and $F_0 \subseteq F_2$, to obtain that $F_1 = F_0 = F_2$. It follows that Ψ is injective and hence there are finitely many fields between K and E .

Let us prove $\impliedby$. Let us assume that $E = K(x, y)$. For each $a \in K$, we consider the extension $K(ay + x)/K$. By assumption, there exist $a, b \in K$ such that $a \neq b$ and

$$K(x + ay) = K(x + by) = L.$$

We claim that $L = E$. Note that $x + ay \in L$ and $x + by \in L$, so $(a - b)y \in L$ and hence, since $K \subseteq L$, it follows that $y \in L$. Thus $x \in L$ and therefore $L = E$. □

§ 9. Galois extensions

As a consequence, if E/K is finite and separable, then E/K admits finitely many subextensions.

§ 9. Galois extensions

Let E/K be an algebraic extension. Assume that $E = K(S)$ and let C be an algebraic closure of K containing E . Let

$$T = \{y \in C : y \text{ is a root of } f(x, K) \text{ for } x \in S\}$$

and let $L = K(T)$. Then $E \subseteq L$, as $S \subseteq T$. The extension L/K is normal, as L/K is a splitting field of the family $\{f(x, K) : x \in S\}$. Moreover, L is the smallest normal extension of K containing E . The field L is the **normal closure** of E (with respect to C).

9.1. EXERCISE. If E/K is finite, then L/K is finite

9.2. EXERCISE. If E/K is separable, then L/K is separable.

Let E/K be an extension and $S \subseteq \text{Gal}(E/K)$ be a subset. The set

$${}^S E = \{x \in E : \sigma(x) = x \text{ for all } \sigma \in S\}$$

is a subfield of E that contains K . The subfield ${}^S E$ is known as the **fixed field** of S .

9.3. DEFINITION. Let E/K be an algebraic extension and $G = \text{Gal}(E/K)$. Then E/K is a **Galois extension** if ${}^G E = K$.

Clearly, K/K is a Galois extension. Note that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not a Galois extension. Why?

9.4. EXERCISE. Prove that $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is a Galois extension.

9.5. EXERCISE. If the characteristic of K is different from two and E/K is an extension of degree 2, then it is Galois.

9.6. EXERCISE. Let E/K be an algebraic extension and $G = \text{Gal}(E/K)$. Let $F = {}^G E$. Prove that $\text{Gal}(E/F) = G$ and hence E/F is a Galois extension.

9.7. PROPOSITION. *Let E/K be an algebraic extension. Then E/K is a Galois extension if and only if E/K is normal and separable.*

PROOF. Let $G = \text{Gal}(E/K)$. Let us first assume that E/K is Galois. For $x \in E$ let

$$f_x = \prod_{y \in O_G(x)} (X - y) = \sum a_i X^i \in E[X].$$

If $\varphi \in G$, then

$$\overline{\varphi}(f_x) = \prod_{y \in O_G(x)} (X - \varphi(y)) = f_x,$$

as if $O_G(x) = \{\sigma_1(x), \dots, \sigma_r(x)\}$, then $\varphi(\sigma_i(x)) = (\varphi\sigma_i)(x) = \sigma_j(x)$ for some j . Since

$$\sum a_i X^i = f_x = \overline{\varphi}(f_x) = \sum \varphi(a_i) X^i,$$

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it follows that $a_i \in {}^G E = K$ for all i . Thus $f_x \in K[X]$ and E/K is a splitting field of the family $\{f_x : x \in E\}$. In particular, E/K is normal. Moreover, x is a simple root of $f_x \in K[X]$ and hence x is separable over K .

Conversely, let $x \in {}^G E$. Since E/K is normal, then $f(x, K) = \prod_{y \in O_G(x)} (X - y)^m$ for some m . Since E/K is separable, $m = 1$. Moreover $x \in {}^G E$, so $O_G(x) = \{x\}$. Thus $f(x, K) = \prod_{y \in O_G(x)} (X - y) = X - x$ and $x \in K$. $\square$

9.8. DEFINITION. Let K be a field and $f \in K[X]$. Then f is **separable** if all roots of f are simple (in some algebraic closure of K).

9.9. PROPOSITION. *Let E/K be a finite extension. Then E/K is a Galois extension if and only if E is a splitting field over K of a separable polynomial $f \in K[X]$.*

PROOF. Let us assume first that E/K is a Galois extension. Since E/K is finite and separable, $E = K(x)$ by Proposition 7.10. Then E/K is a splitting field of $f(x, K)$ since E/K is normal. Since E/K is separable, x is separable over K . Thus x is a simple root of $f(x, K)$ and hence $f(x, K)$ is separable. Conversely, let $x_1, \dots, x_r$ be the roots of a separable polynomial $f \in K[X]$. Then $E = K(x_1, \dots, x_r)$ is separable and normal. $\square$

In the previous case, $\text{Gal}(E/K)$ is known as the **Galois group** of the polynomial f . The notation is $\text{Gal}(f, K)$. If $n = \deg f$ and $x_1, \dots, x_n$ are the roots of f , then any $\varphi \in \text{Gal}(f, K)$ permutes the roots of f , that is φ permutes the set $\{x_1, \dots, x_n\}$. In particular, $\text{Gal}(f, K)$ is isomorphic to a subgroup of $\mathbb{S}_n$ and hence $|\text{Gal}(f, K)|$ divides $n!$.

9.10. PROPOSITION. *Let E/K be a normal extension and F be the separable closure of K with respect to E . Then F/K is a Galois extension.*

PROOF. Let C/K be an algebraic closure such that $E \subseteq C$. Let $\sigma \in \text{Hom}(F/K, C/K)$, and let $\varphi \in \text{Hom}(E/K, C/K)$ be such that $\varphi|_F = \sigma$. Since E/K is normal, $\varphi(E) = E$. Let $x \in F$. Then $\sigma(x) = \varphi(x) \in E$. Thus $f(\sigma(x), K) = f(x, K)$ and $\sigma(x)$ is separable over K , which implies that $\sigma(x) \in F$. Thus F/K is normal. Since F/K is separable, it follows that F/K is a Galois extension by Proposition 9.7. $\square$

Some easy facts.

9.11. EXERCISE. Let E/K be a separable extension and L/K be the normal closure of E in some algebraic closure C that contains E . Prove that L/K is a Galois extension.

9.12. EXERCISE. Let E/K be a finite extension. Prove that E/K is Galois if and only if $[E : K] = |\text{Gal}(E/K)|$.

For the previous exercise, note that if E/K is a finite extension, then

$$|\text{Gal}(E/K)| \leq \gamma(E/K) \leq [E : K].$$

The first inequality is equality if and only if E/K is normal. The second inequality is equality if and only if E/K is separable.

9.13. EXERCISE. Let E/K be a Galois extension and F/K be a subextension of E/K . Prove that E/F is a Galois extension.

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9.14. THEOREM (Artin). *Let E be a field and G be a finite group of automorphisms of E . If $K = {}^G E$, then E/K is a Galois extension, $[E : K] = |G|$ and $\text{Gal}(E/K) = G$.*

Before proving the theorem, we need a lemma.

9.15. LEMMA. *Let E/K be a separable extension such that $\deg f(x, K) \leq m$ for all $x \in E$. Then E/K is finite and $[E : K] \leq m$.*

PROOF. Let $z \in E$ be of maximal degree. If $x \in E$, then $K(x, z)/K$ is separable. There exists y such that $K(x, z) = K(y)$. Then

$$K(z) \subseteq K(x, z) = K(y).$$

Since $\deg f(z, K) \leq \deg f(y, K)$, $\deg f(z, K) = \deg f(y, K)$. Hence $K(y) = K(z)$. In particular, $x \in K(z)$ and therefore $E = K(z)$. $\square$

Now we are ready to prove Artin's theorem:

PROOF OF THEOREM 9.14. Note that $G \subseteq \text{Gal}(E/K)$. Let $x \in E$ and

$$f_x = \prod_{y \in O_G(x)} (X - y).$$

Since $f_x \in K[X]$, the extension E/K is normal and separable (as it is a splitting field of a family of separable polynomials), so E/K is a Galois extension. Moreover,

$$\deg f(x, K) \leq \deg f_x = |O_G(x)| \leq |G|.$$

By the previous lemma, E/K is finite and $[E : K] \leq |G|$. This implies that

$$|\text{Gal}(E/K)| = [E : K] \leq |G|$$

and hence $|\text{Gal}(E/K)| = |G|$. $\square$

9.16. EXAMPLE. Let $E = K(X, Y)$ and $\sigma: K[X, Y] \rightarrow E$ be the ring homomorphism given by $\sigma(X) = Y$ and $\sigma(Y) = X$. Note that σ is bijective, as $\sigma^2 = \text{id}$. The map σ induces a field homomorphism $\bar{\sigma}: E \rightarrow E$ such that $\bar{\sigma}^2 = \text{id}$. Recall that such a homomorphism is given by $f/g \mapsto \sigma(f)/\sigma(g)$. Let $G = \langle \bar{\sigma} \rangle$. Then $|G| = 2$. We claim that ${}^G E = K(X+Y, XY)$. Let $F = K(X+Y, XY)$. We only prove that ${}^G E \subseteq F$, as the other inclusion is trivial. Artin's theorem implies that $[E : {}^G E] = 2$ and $E = F(X)$, as X is a root of the polynomial $Z^2 - (X+Y)Z + XY$. Then $[E : F] \leq 2$ and $[{}^G E : F] = 1$.

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A **partially order set** (or **poset**) is a pair $(X, \leq)$, where X is a non-empty set and $\leq$ is a reflexive, antisymmetric and transitive relation on X . This means:

- 1) $x \leq x$ for all $x \in X$,
- 2) $x \leq y$ and $y \leq x$ imply $x = y$, and
- 3) $x \leq y$ and $y \leq z$ imply $x \leq z$.

Let $(X, \leq)$ be a partially ordered set and $x, y \in X$. An element z of a poset $(X, \leq)$ is an **upper bound** of x and y if $x \leq z$ and $y \leq z$. And ξ is a **least upper bound** of x and y if it is an upper bound with $\xi \leq z$ for every upper bound z of x and y . Similarly, one defines lower bounds and greatest lower bounds.

10.1. DEFINITION. A **lattice** is a partially ordered set $\mathcal{L}$ in which each pair of elements $x, y \in \mathcal{L}$ has a least upper bound $x \vee y$ and a greatest lower bound $x \wedge y$.

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The basic example is the following. Let X be a set and $\mathcal{P}(X)$ be the collection of all subsets of X . The relation $A \leq B \iff A \subseteq B$ turns $\mathcal{P}(X)$ into a lattice with $A \vee B = A \cup B$ and $A \wedge B = A \cap B$.

10.2. EXAMPLE. Let G be a group and $\mathcal{L}(G)$ be the collection of subgroups of G . The relation

$$H \leq K \iff H \subseteq K$$

turns $\mathcal{L}(G)$ into a lattice with $H \vee K = \langle H, K \rangle$ and $H \wedge K = H \cap K$.

10.3. EXAMPLE. Let E/K be a field extension and $\mathcal{L}(E/K)$ be the collection of intermediate fields. The relation

$$F \leq L \iff F \subseteq L$$

turns $\mathcal{L}(E/K)$ into a lattice with $F \vee L = FL$ and $F \wedge L = F \cap L$.

10.4. EXAMPLE. Let $n \in \mathbb{Z}$ be a positive integer and $\mathcal{D}(n)$ be the collection of positive integer divisors $d \in \mathbb{Z}$ of n . The relation

$$l \leq m \iff l \mid m$$

turns $\mathcal{D}(n)$ into a lattice with $l \vee m = \text{lcm}(l, m)$ and $l \wedge m = \text{gcd}(l, m)$.

A map $f: \mathcal{L}_1 \rightarrow \mathcal{L}_2$ between two lattices is said to be **order-reversing** if $x \leq_1 y$ implies $f(y) \leq_2 f(x)$.

We shall need an exercise.

10.5. EXERCISE. Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be lattices and $f: \mathcal{L}_1 \rightarrow \mathcal{L}_2$ be a bijection such that f and its inverse are both order reversing. Then

$$f(x \vee y) = f(x) \wedge f(y), \quad f(x \wedge y) = f(x) \vee f(y)$$

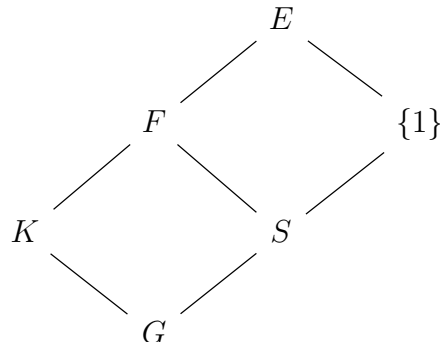
for all $x, y \in \mathcal{L}_1$.

10.6. THEOREM (Galois). Let E/K be a finite Galois extension and $G = \text{Gal}(E/K)$. There exists a bijective correspondence

$$\mathcal{L}(E/K) = \{F : K \subseteq F \subseteq E \text{ subfields}\} \leftrightarrow \{S : S \text{ is a subgroup of } G\} = \mathcal{L}(G).$$

The correspondence is given by $\alpha: F \mapsto \text{Gal}(E/F)$ and $\beta: {}^S E \mapsto S$. Moreover, the following conditions hold:

- 1) α and β are order-reversing bijections.
- 2) $[F : K] = (G : \text{Gal}(E/F))$ and $(G : S) = [{}^S E : K]$.
- 3) F/K is a Galois extension if and only if $\text{Gal}(E/F)$ is a normal subgroup of G .



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PROOF. Let

$$\begin{aligned}\alpha: \mathcal{L}(E/K) &\rightarrow \mathcal{L}(G), & \alpha(F) &= \text{Gal}(E/F), \\ \beta: \mathcal{L}(G) &\rightarrow \mathcal{L}(E/K), & \beta(S) &= {}^S E.\end{aligned}$$

A routine exercise shows that α and β are well-defined. We first note that

$$\beta(\alpha(F)) = \beta(\text{Gal}(E/F)) = {}^{\text{Gal}(E/F)} E = F$$

since E/F is a Galois extension. Moreover,

$$\alpha(\beta(S)) = \alpha({}^S E) = \text{Gal}(E/{}^S E) = S$$

by Artin's theorem, as S is finite.

It is straightforward to check that α and β are order-reversing bijections.

Let F be a subfield of E containing K and $S = \alpha(F)$. Then

$$[F : K] = \frac{[E : K]}{[E : F]} = \frac{|G|}{|S|} = (G : S).$$

Let C be an algebraic closure of K that contains E . If $S = \text{Gal}(E/F)$, then $F = {}^S E$.

We need to prove that F/K is normal if and only if S is normal in G . Let us first prove $\implies$. Let $\tau \in S$ and $\sigma \in G$. Since F/K is normal, $\sigma|_F \in \text{Aut}(F)$. Thus $\sigma^{-1}(F) = F$. In particular, if $x \in F$, then $\sigma^{-1}(x) \in F$ and

$$\sigma\tau\sigma^{-1}(x) = \sigma\sigma^{-1}(x) = x.$$

Conversely, let $\varphi \in \text{Hom}(F/K, C/K)$. There exists $\Phi: E \rightarrow C$ such that $\Phi|_F = \varphi$. Since E/K is normal, $\Phi(E) = E$ and hence $\Phi \in G$. We claim that $\varphi(x) \in F$ for all $x \in F$. Note that $F = {}^S E$, so

$$\tau\varphi(x) = \tau\Phi(x) = \Phi\Phi^{-1}\tau\Phi(x) = \Phi(x) = \varphi(x)$$

for all $\tau \in S$, as $\Phi^{-1}\tau\Phi \in S$. This means that $\varphi(x) \in {}^S E = F$.

Let us compute $\text{Gal}(F/K)$. Since F/K is normal, the map $\lambda: G \rightarrow \text{Gal}(F/K)$, $\sigma \mapsto \sigma|_F$, is a surjective group homomorphism such that $\ker \lambda = S$. The first isomorphism theorem implies that $\text{Gal}(F/K) \simeq G/S$. $\square$

Some easy consequences.

10.7. EXERCISE. If E/K is a Galois extension of degree n and p is a prime number dividing n , then E/K admits a subextension of degree n/p .

10.8. EXERCISE. If E/K is a Galois extension of degree $p^\alpha m$ with p a prime number coprime with m , then E/K admits a subextension of degree m .

10.9. DEFINITION. An extension E/K is **abelian** if E/K is a Galois extension with $\text{Gal}(E/K)$ abelian.

10.10. EXERCISE. If E/K is an abelian extension of degree n and d divides n , then E/K admits a subextension of degree d .

10.11. DEFINITION. An extension E/K is **cyclic** if E/K is a Galois extension with $\text{Gal}(E/K)$ cyclic.

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10.12. EXAMPLE. The extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ admits exactly three non-trivial subextensions:

$$\mathbb{Q}(\sqrt{2})/\mathbb{Q}, \quad \mathbb{Q}(\sqrt{3})/\mathbb{Q}, \quad \mathbb{Q}(\sqrt{6})/\mathbb{Q},$$

as $\text{Gal}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}) \simeq C_2 \times C_2$.

10.13. EXAMPLE. Let $\omega \in \mathbb{C} \setminus \{1\}$ be such that $\omega^5 = 1$. Then

$$f(\omega, \mathbb{Q}) = 1 + X + X^2 + X^3 + X^4$$

and $\mathbb{Q}(\omega)/\mathbb{Q}$ has degree four. Moreover, $\mathbb{Q}(\omega)/\mathbb{Q}$ is a Galois extension and

$$\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \simeq C_4.$$

If $\sigma \in \text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$, then $\sigma(\omega) = \omega^i$ for some $i \in \{1, \dots, 4\}$. Moreover, each map $\omega \mapsto \omega^i$, for $i \in \{1, \dots, 4\}$, induces an automorphism of $\mathbb{Q}(\omega)/\mathbb{Q}$. Thus $|\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})| = 4$. Now

$$\sigma_i^k = \text{id} \iff \omega^{i^k} = \sigma_i^k(\omega) = \omega \iff i^k \equiv 1 \pmod{5}.$$

Thus the map σ_2 given by $\omega \mapsto \omega^2$ has order four.

Since $\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) = \langle \sigma \rangle$, where $\sigma(\omega) = \omega^2$, is cyclic of order four, the extension $\mathbb{Q}(\omega)/\mathbb{Q}$ has a unique degree-two subextension $F/\mathbb{Q}$. Note that $|\langle \sigma^2 \rangle| = 2$ and $\sigma^2(\omega) = \omega^4 = \omega^{-1}$. Thus $F = \langle \sigma^2 \rangle \mathbb{Q}(\omega)$. Let $\theta = \omega + \omega^{-1}$. Then

$$\theta^2 = \omega^2 + \omega^3 + 2 = -(1 + \omega + \omega^{-1}) + 2 = 1 - \theta$$

and hence θ is a root of $X^2 + X - 1$. It follows that

$$\theta \in \{(-1 + \sqrt{5})/2, (-1 - \sqrt{5})/2\}.$$

Therefore $F = \mathbb{Q}(\sqrt{5})$.

Let us mention some other consequences (of the fact that the correspondence depends on order-reversing bijections).

10.14. EXERCISE. Let E/K be a finite Galois extension and $G = \text{Gal}(E/K)$. If S and T are subgroups of G , then $\langle S, T \rangle E = {}^S E \cap {}^T E$ and ${}^{S \cap T} E = {}^S E {}^T E$.

10.15. EXERCISE. Let E/K be a finite Galois extension and $F, L \in \mathcal{L}(E/K)$. Prove that $\text{Gal}(E/FL) = \text{Gal}(E/F) \cap \text{Gal}(E/L)$ and $\text{Gal}(E/F \cap L) = \langle \text{Gal}(E/F), \text{Gal}(E/L) \rangle$.

10.16. EXERCISE. Let E/K be a finite Galois extension and $G = \text{Gal}(E/K)$. Assume that G is the direct product $G = S \times T$ of the groups S and T . Let $F = {}^S E$ and $L = {}^T E$. Then $F \cap L = K$ and $FL = E$.

10.17. PROPOSITION. Let $E_1/K, E_2/K$ be Galois extensions. If $E = E_1 E_2$, then E/K is a Galois extension. If, moreover, E_1/K and E_2/K are finite, then

$$\theta: \text{Gal}(E/K) \rightarrow \text{Gal}(E_1/K) \times \text{Gal}(E_2/K), \quad \sigma \mapsto (\sigma|_{E_1}, \sigma|_{E_2}),$$

is an injective group homomorphism.

§ 11. The fundamental theorem of algebra

PROOF. Since E_1/K is algebraic, E_1E_2/E_2 is algebraic. Since E_2/K is algebraic, E_1E_2/K is algebraic. Similarly, E_1E_2/K is separable.

Let C/K be an algebraic closure such that $E_1E_2 \subseteq C$. If $\sigma \in \text{Hom}(E_1E_2/K, C/K)$, then $\sigma(E_1E_2) \subseteq \sigma(E_1)\sigma(E_2) = E_1E_2$ (do this calculation as an exercise using the fact that E_1/K and E_2/K are normal extensions). Thus E_1E_2/K is normal.

If both E_1/K and E_2/K are finite, then E_1E_2/K is finite.

Then θ is a group homomorphism. We claim that the map θ is injective. Let $\sigma \in \ker \theta$. Then $\sigma|_{E_i} = \text{id}_{E_i}$ for all $i \in \{1, 2\}$. Let $S = \langle \sigma \rangle \subseteq \text{Gal}(E/K)$ and $F = {}^SE$. Then $E_i \subseteq F$ for all $i \in \{1, 2\}$ and hence $E \subseteq F$. It follows that $F = E = {}^{\{\text{id}\}}E$ and therefore $S = \{\text{id}\}$, so $\sigma = \text{id}$. $\square$

10.18. EXERCISE. Let $E_1/K, \dots, E_r/K$ be finite Galois extensions such that for each j one has $E_j \cap (E_1 \cdots E_{j-1}E_{j+1} \cdots E_r) = K$. If $E = E_1E_2 \cdots E_r$, then

$$\text{Gal}(E/K) \simeq \text{Gal}(E_1/K) \times \cdots \times \text{Gal}(E_r/K).$$

In this case, $[E : K] = \prod_{i=1}^r [E_i : K]$.

§ 11. The fundamental theorem of algebra

We now present an easy proof of the fundamental theorem of algebra based on the ideas of Galois Theory. We need the following well-known facts:

- 1) Every real polynomial of odd degree admits a real root. This means that $\mathbb{R}$ does not admit extensions of odd degree > 1 .
- 2) Every complex number admits a square root in $\mathbb{C}$. This means that $\mathbb{C}$ does not admit degree-two extensions.

11.1. THEOREM. *The field $\mathbb{C}$ is algebraically closed.*

PROOF. Let $E/\mathbb{C}$ be an algebraic finite extension. Then $E/\mathbb{R}$ is a finite separable extension of even degree. There exists a Galois extension $L/\mathbb{R}$ such that $E \subseteq L$, so $[L : \mathbb{R}]$ is even. Let $G = \text{Gal}(L/\mathbb{R})$. Then $|G| = 2^m s$ for some odd number s . If T is a 2-Sylow subgroup of G , then there exists a subextension $F/\mathbb{R}$ of degree s . Since $\mathbb{R}$ does not admit extensions of odd degree > 1 , $s = 1$ and hence G is a 2-group. Since $L/\mathbb{R}$ is a Galois extension, $L/\mathbb{C}$ is a Galois extension. In particular, $|\text{Gal}(L/\mathbb{C})| = 2^{m-1}$. If $m > 1$, let U be a subgroup of $\text{Gal}(L/\mathbb{C})$ of order 2^{m-2} . Then U corresponds to a subextension $L_1/\mathbb{C}$ of degree two, a contradiction. Hence $m = 1$ and $[L : \mathbb{C}] = 1$, so $L = \mathbb{C}$ and $E = \mathbb{C}$. $\square$

§ 12. Purely inseparable extensions

Let E/K be an algebraic extension. In Chapter 7 we defined the **separable closure** of K with respect to E as the field

$$F = \{x \in E : x \text{ is separable over } K\}.$$

Note that $K \subseteq F \subseteq E$ and $F = K(F)$. Moreover, F/K is separable and E/F is a **purely inseparable** extension, meaning that for every $x \in E \setminus F$, the polynomial $f(x, F)$ is not separable.

The number $[E : K]_{\text{ins}} = [E : F]$ is known as the **degree of inseparability** of E/K . Clearly, E/K is separable if and only if $[E : K]_{\text{ins}} = 1$ and E/K is purely inseparable if and only if $[E : K]_{\text{ins}} = [E : K]$.

§ 12. Purely inseparable extensions

12.1. EXERCISE. Let K be a field of characteristic $p > 0$ and $f \in K[X]$ be irreducible. If f is not separable, then $f = g(X^p)$ for some $g \in K[X]$.

12.2. PROPOSITION. Let K be a field of characteristic $p > 0$ and E/K be an algebraic extension. The following statements are equivalent:

- 1) E/K is purely inseparable.
- 2) If $x \in E$, then $x^{p^m} \in K$ for some $m \geq 0$.
- 3) If $x \in E$, then $f(x, K) = X^{p^m} - a$ for some $a \in K$ and $m \geq 0$.
- 4) $\gamma(E/K) = 1$.

PROOF. We first prove 1) $\implies$ 2). Let $x \in E$ and $f = f(x, K)$. Assume x is not separable. Then $f(x) = 0$ and $f'(x) = 0$, as x is not a simple root. By Exercise 12.1, $f = g(X^p)$ for some $g \in K[X]$. We now proceed by induction on the degree of x . The result is true for elements of degree one. So assume the result holds for the element of degree $\leq n$ for some $n \geq 1$. If $x \in E$ is such that $\deg f(x, K) = n + 1$, then, since $f(x, K) = g(X^p)$, the element x^p has degree $\leq n$. By the inductive hypothesis, $x^{p^{m+1}} = (x^p)^{p^m} \in K$.

We now prove 2) $\implies$ 3). Let $x \in E$ and m be the minimal positive integer such that $x^{p^m} \in K$. Then x is a root of $X^{p^m} - x^{p^m} \in K[X]$. Since $X^{p^m} - x^{p^m} = (X - x)^{p^m}$, it follows that

$$f(x, K) = (X - x)^r = X^r + \cdots + (-1)^r x^r$$

for some $r \in \{1, \dots, p^m\}$. Write $r = p^s t$ for some integer t coprime with p and s such that $0 \leq s \leq m$. Let $a, b \in \mathbb{Z}$ be such that $ar + bp^m = p^s$. Then

$$x^{p^s} = x^{ar+bp^m} = (x^r)^a (x^{p^m})^b \in K.$$

The minimality of m implies that $s \geq m$ and hence $s = m$. Now $p^m t = p^s t = r \leq p^m$, so $t = 1$. This means $f(x, K) = X^{p^m} - x^{p^m}$.

We now prove 3) $\implies$ 4). Let C/K be an algebraic closure of K containing E and $x \in E$. Let $\sigma \in \text{Hom}(E/K, C/K)$. We claim that $\sigma(x) = x$. Since $f(x, K) = X^{p^m} - a$,

$$(\sigma(x))^{p^m} = \sigma(x^{p^m}) = \sigma(a) = a = x^{p^m}.$$

It follows that $\sigma(x)$ is a root of $X^{p^m} - x^{p^m} = (X - x)^{p^m}$. Thus $\sigma(x) = x$.

Finally, we prove that 4) $\implies$ 1). Let C be an algebraic closure of K containing E . Then $\text{Gal}(E/K) \subseteq \text{Hom}(E/K, C/K) = \{\text{id}\}$, as $\gamma(E/K) = 1$. If $x \in E$ is separable over K , then

$$f(x, K) = \prod_{y \in O_{\text{Gal}(E/K)}(x)} (X - y) = X - x \in K[X].$$

Thus $x \in K$ and hence E/K is purely inseparable. □

Some consequences:

12.3. EXERCISE. Let K be a field of characteristic $p > 0$ and E/K be finite and purely inseparable. Then $[E : K] = p^s$ for some prime number p and some integer s . Moreover, for any $x \in E$, we have that $x^{[E:K]} \in K$.

For the first part of the previous exercise, write $E = K(x_1, \dots, x_n)$ and proceed by induction on n .

§ 13. Norm and trace

12.4. EXERCISE. Let K be of characteristic $p > 0$ and E/K be a finite extension such that $[E : K]$ is not divisible by p . Then E/K is separable.

Let K be of characteristic $p > 0$, E/K be finite and F be the separable closure of K in E . Since

$$\gamma(E/K) = \gamma(E/F)\gamma(F/K) = \gamma(F/K),$$

it follows that

$$[E : K] = [E : F]\gamma(E/K) = [E : K]_{\text{ins}}\gamma(E/K).$$

§ 13. Norm and trace

13.1. DEFINITION. Let E/K be a finite extension and C/K be an algebraic closure that contains E . Let $A = \text{Hom}(E/K, C/K)$. For $x \in E$ we define the **trace** of x in E/K as

$$\text{trace}_{E/K}(x) = [E : K]_{\text{ins}} \sum_{\varphi \in A} \varphi(x)$$

and the **norm** of x in E/K as

$$\text{norm}_{E/K}(x) = \left(\prod_{\varphi \in A} \varphi(x) \right)^{[E:K]_{\text{ins}}}.$$

As an optional exercise, one can show that these definitions do not depend on the algebraic closure.

We collect some basic properties as an exercise:

13.2. EXERCISE. Let E/K be a finite extension. The following statements hold:

- 1) If E/K is not separable, then $\text{trace}_{E/K}(x) = 0$ for all $x \in E$.
- 2) If $x \in K$, then $\text{trace}_{E/K}(x) = [E : K]x$.
- 3) $\text{trace}_{E/K}(x) \in K$ for all $x \in E$.
- 4) $\text{norm}_{E/K}(x) = 0$ if and only if $x = 0$.
- 5) If $x \in K$, then $\text{norm}_{E/K}(x) = x^{[E:K]}$.
- 6) $\text{norm}_{E/K}(x) \in K$ for all $x \in E$.

One proves, moreover, that $\text{trace}_{E/K}: E \rightarrow K$ satisfies

$$\text{trace}_{E/K}(x + \lambda y) = \text{trace}_{E/K}(x) + \lambda \text{trace}_{E/K}(y)$$

for all $x, y \in E$ and $\lambda \in K$, that is to say that $\text{trace}_{E/K}: E \rightarrow K$ is a linear form in E . Furthermore $\text{norm}_{E/K}: E^\times \rightarrow K^\times$ is a group homomorphism.

13.3. EXERCISE. Let E/K be a finite extension and $x \in E$ be algebraic. If

$$f(x, K) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0,$$

then $\text{norm}_{E/K}(x) = ((-1)^n a_0)^{[E:K(x)]}$ and $\text{trace}_{E/K}(x) = -[E : K(x)]a_{n-1}$.

§ 14. Finite fields

13.4. EXAMPLE. Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Then

$$\begin{aligned} \text{trace}_{E/\mathbb{Q}}(\sqrt{2}) &= 0, & \text{norm}_{E/\mathbb{Q}}(\sqrt{2}) &= 4, \\ \text{trace}_{E/\mathbb{Q}(\sqrt{2})}(\sqrt{2}) &= 2\sqrt{2}, & \text{norm}_{E/\mathbb{Q}(\sqrt{2})}(\sqrt{2}) &= 2. \end{aligned}$$

13.5. EXAMPLE. If E/K is a finite Galois extension, then

$$\text{trace}_{E/K}(x) = \sum_{\sigma \in \text{Gal}(E/K)} \sigma(x) \quad \text{and} \quad \text{norm}_{E/K}(x) = \prod_{\sigma \in \text{Gal}(E/K)} \sigma(x)$$

for all $x \in E$. In particular, since $E = K(y)$ for some y by Proposition 7.10,

$$\text{trace}_{E/K}(y) = -a_{n-1} \quad \text{and} \quad \text{norm}_{E/K}(y) = (-1)^n a_0,$$

where $f(y, K) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0$.

§ 14. Finite fields

In this section, p will be a prime number.

14.1. PROPOSITION. *Let m be a positive integer. Up to isomorphism, there exists a unique field F_{p^m} of size p^m .*

PROOF. Let C be an algebraic closure of the field $\mathbb{Z}/p$ and let $F_{p^m} = \{x \in C : x^{p^m} = x\}$ be the set of roots of $X^{p^m} - X$. Since the polynomial $X^{p^m} - X$ has no multiple roots, $|F_{p^m}| = p^m$. Moreover, F_{p^m} is the unique subfield of C of size p^m .

To prove the uniqueness, it is enough to note that if K is a field of p^m elements, then K is the splitting field of $X^{p^m} - X$ over $\mathbb{Z}/p$. $\square$

Let $K = \mathbb{Z}/p$ and C be an algebraic closure of K . We claim that $C = \bigcup_k F_{p^k}$. If $x \in C$, then x is algebraic over K . Since $K(x)/K$ is finite, $K(x)$ is a finite field, say $|K(x)| = p^r$ for some r . Then $x^{p^r} = x$ and hence $x \in F_{p^r}$.

14.2. EXERCISE. Prove the following statements:

- 1) If $x \in F_{p^r}$, then $x^{p^{rk}} = x$ for all $k \geq 0$.
- 2) $F_{p^m} \subseteq F_{p^n}$ if and only if $m \mid n$.
- 3) $F_{p^m} \cap F_{p^n} = F_{p^{\gcd(m,n)}}$.

14.3. PROPOSITION. *Every finite extension of a finite field is cyclic.*

PROOF. Let $K = \mathbb{Z}/p$. It is enough to show that F_{p^n}/F_{p^m} is cyclic if m divides n . We first prove that F_{p^n}/K is cyclic. Let

$$\sigma: F_{p^n} \rightarrow F_{p^n}, \quad x \mapsto x^p.$$

Then $\sigma \in \text{Gal}(F_{p^n}/K)$ (it is bijective because all field homomorphisms are injective and F_{p^n} is finite).

Note that F_{p^n}/K is a Galois extension, as F_{p^n} is the splitting field over K of the separable polynomial $X^{p^n} - X \in K[X]$. Thus $|\text{Gal}(F_{p^n}/K)| = [F_{p^n} : K] = n$.

We claim that σ generates $\text{Gal}(F_{p^n}/K)$. Since $\sigma^i(x) = x^{p^i}$ for all $i \geq 0$, in particular,

$$\sigma^n(x) = x^{p^n} = x.$$

§ 15. Cyclotomic extensions

Thus $\sigma^n = \text{id}$ and hence $|\sigma|$ divides n . Let $s = |\sigma|$. We know that $F_{p^n}^\times = F_{p^n} \setminus \{0\}$ is cyclic, say $F_{p^n}^\times = \langle g \rangle$. Since $|g| = p^n - 1$,

$$g = \sigma^s(g) = g^{p^s}$$

and hence $p^s \equiv 1 \pmod{p^n - 1}$. Thus $p^n - 1$ divides $p^s - 1$ and hence n divides s . Therefore $n = s$ and $\text{Gal}(F_{p^n}/K) = \langle \sigma \rangle$.

For the general case, note that if m divides n , then the Galois group $\text{Gal}(F_{p^n}/F_{p^m})$ is a subgroup of $\text{Gal}(F_{p^n}/K)$. Since $\text{Gal}(F_{p^n}/K)$ is cyclic, the claim follows. $\square$

If $K = \mathbb{Z}/p$ and m divides n , the subextension F_{p^m} corresponds to the unique subgroup of index m of $\text{Gal}(F_{p^n}/K) = \langle \sigma \rangle$. This subgroup is $\langle \sigma^m \rangle$, where for all $x \in F_{p^n}$,

$$\sigma^m(x) = x^{p^m} = x^{|F_{p^m}|}.$$

Note that $\text{Gal}(F_{p^n}/F_{p^m}) = \langle \sigma^m \rangle$. The map σ^m is known as the **Frobenius automorphism**.

14.4. EXERCISE. Let E/K be an extension of finite fields. Then E/K is cyclic. Moreover, $\text{Gal}(E/K) = \langle \tau \rangle$, where $\tau(x) = x^{|K|}$.

§ 15. Cyclotomic extensions

For $n \geq 1$ let $G_n(K) = \{x \in K : x^n = 1\}$ be the set of n -roots of one in K . Note that $G_n(K)$ is a cyclic subgroup of $K^\times$ and that $|G_n(K)|$ divides n .

15.1. EXAMPLE. $G_n(\mathbb{R}) = \{-1, 1\}$ if n is even and $G_n(\mathbb{R}) = \{1\}$ if n is odd.

15.2. EXERCISE. Let K be a field of characteristic $p > 0$. Let $n = p^s m$ for some m not divisible by p . Then $G_n(K) = G_m(K)$.

15.3. EXERCISE. Let q be a prime number. Then $G_n(\mathbb{Z}/q) \simeq \mathbb{Z}/\gcd(n, q-1)$.

Similarly, one can prove that if K is a finite field, then $G_n(K)$ is a cyclic group of order $\gcd(n, |K^\times|)$.

15.4. EXAMPLE. If C is algebraically closed of characteristic coprime with n , then $G_n(C)$ is cyclic of order n , as $X^n - 1$ has all its roots in C and does not contain multiple roots.

Let K be an algebraically closed field and n be such that n is coprime with the characteristic of K . The set of **primitive n -roots** is defined as

$$H_n(K) = \{x \in G_n(K) : |x| = n\}.$$

15.5. DEFINITION. Let K be an algebraically closed field and n be such that n is coprime with the characteristic of K . The **n -th cyclotomic polynomial** is defined as

$$\Phi_n = \prod_{x \in H_n(K)} (X - x) \in K[X].$$

For $n \geq 1$ the Euler's function is defined as

$$\varphi(n) = |\{k : 1 \leq k \leq n, \gcd(k, n) = 1\}|.$$

For example, $\varphi(4) = 2$, $\varphi(8) = \varphi(10) = 4$ and $\varphi(p) = p - 1$ for every prime p .

§ 15. Cyclotomic extensions

15.6. PROPOSITION. Let K be an algebraically closed field and n be such that n is coprime with the characteristic of K . Let A be the prime subring of K .

- 1) $\deg \Phi_n = \varphi(n)$.
- 2) $\Phi_n \in A[X]$.

PROOF. The first statement is clear. Let us prove 2) by induction on n . The case $n = 1$ is trivial, as $\Phi_1 = X - 1$. Assume that $\Phi_d \in A[X]$ for all d such that $d < n$. In particular,

$$\gamma = \prod_{\substack{d|n \\ d \neq n}} \Phi_d \in A[X].$$

Since γ is monic, it follows that $\frac{X^n - 1}{\gamma} \in A[X]$. Now the claim follows from

$$X^n - 1 = \prod_{d|n} \Phi_d = \Phi_n \prod_{\substack{d|n \\ d \neq n}} \Phi_d = \Phi_n \gamma. \quad \square$$

By taking degree in the equality $X^n - 1 = \prod_{d|n} \Phi_d$ one gets

$$n = \sum_{d|n} \varphi(d).$$

15.7. DEFINITION. Let $n \geq 2$ and K be a field of characteristic coprime with n . A **cyclotomic extension** of K of index n is a splitting field of $X^n - 1$ over K .

Let C be an algebraic closure of K and $n \geq 2$ be coprime with the characteristic of K . It follows from Definition 15.7 that a cyclotomic extension of K of index n is of the form $K(\omega)/K$ for some $\omega \in H_n(C)$.

15.8. PROPOSITION. A cyclotomic extension of index n is abelian and of degree a divisor of $\varphi(n)$.

PROOF. Let C be an algebraic closure of K and $n \geq 2$ be coprime with the characteristic of K . Let $\omega \in H_n(C)$ and $K(\omega)/K$ be a cyclotomic extension. Then $K(\omega)/K$ is a Galois extension, as it is a splitting field of a separable polynomial. Let $U = \mathcal{U}(\mathbb{Z}/n)$ be the group of units of $\mathbb{Z}/n$ and

$$\lambda: \text{Gal}(K(\omega)/K) \rightarrow U, \quad \sigma \mapsto m_\sigma,$$

where m_σ is such that $\sigma(\omega) = \omega^{m_\sigma}$. The map λ is well-defined and it is a group homomorphism, as if $\sigma, \tau \in \text{Gal}(K(\omega)/K)$, then, since

$$(\tau\sigma)(\omega) = \tau(\sigma(\omega)) = \tau(\omega^{m_\sigma}) = \tau(\omega)^{m_\sigma} = (\omega^{m_\tau})^{m_\sigma} = \omega^{m_\sigma m_\tau},$$

it follows that $\lambda(\sigma)\lambda(\tau) = \lambda(\sigma\tau)$. Since λ is injective, $\text{Gal}(K(\omega)/K)$ is isomorphic to a subgroup of the abelian group U . Hence $\text{Gal}(K(\omega)/K)$ is abelian. Moreover,

$$[K(\omega) : K] = |\text{Gal}(K(\omega)/K)|$$

is a divisor of $|U| = \varphi(n)$. □

15.9. EXERCISE. Prove that a cyclotomic extension $K(\omega)/K$ has degree $\varphi(n)$ if and only if Φ_n is irreducible over K .

Note that Φ_n is irreducible over $\mathbb{Q}$. Some concrete examples:

$$\Phi_1 = X - 1, \quad \Phi_2 = X + 1, \quad \Phi_3 = X^2 + X + 1, \quad \Phi_6 = X^2 - X + 1.$$

§ 16. Hilbert's theorem 90

If p is a prime number, then $\Phi_p = X^{p-1} + \cdots + X + 1$.

15.10. EXAMPLE. Φ_5 is irreducible over $\mathbb{Z}/2$. First note that $\Phi_5 = X^4 + \cdots + X + 1$ does not have roots in $\mathbb{Z}/2$. If Φ_5 is reducible, then, since $X^2 + X + 1$ is the unique degree-two monic irreducible polynomial over $\mathbb{Z}/2$, it follows that

$$\Phi_5 = (X^2 + X + 1)(X^2 + X + 1) = (X^2 + X + 1)^2 = X^4 + X^2 + 1,$$

a contradiction.

15.11. EXERCISE. Prove that $\Phi_{12} = X^4 - X^2 + 1$ is not irreducible over $\mathbb{Z}/5$.

§ 16. Hilbert's theorem 90

16.1. THEOREM (Hilbert). *Let E/K be a cyclic extension. Assume that $\text{Gal}(E/K)$ is generated by τ . For $a \in E$, $\text{norm}_{E/K}(a) = 1$ if and only if $a = b/\tau(b)$ for some $b \in E \setminus \{0\}$.*

PROOF. Let $n = |\text{Gal}(E/K)|$. We first prove $\Leftarrow$. If $a = b/\tau(b)$ and $b \neq 0$, then

$$\text{norm}_{E/K}(a) = a\tau(a)\tau^2(a)\cdots\tau^{n-1}(a) = \frac{b}{\tau(b)} \frac{\tau(b)}{\tau^2(b)} \cdots \frac{\tau^{n-1}(b)}{\tau^n(b)} = 1.$$

Now we prove $\Rightarrow$. Let $a \in E$ be such that $\text{norm}_{E/K}(a) = 1$. For $c \in E$ let

$$\begin{aligned} d_0 &= ac, \\ d_1 &= a\tau(a)\tau(c), \\ d_2 &= a\tau(a)\tau^2(a)\tau^2(c), \\ &\vdots \\ d_{n-1} &= \underbrace{a\tau(a)\cdots\tau^{n-1}(a)}_{=\text{norm}_{E/K}(a)} \tau^{n-1}(c) = \tau^{n-1}(c). \end{aligned}$$

Then

$$a\tau(d_j) = a\tau(a)\cdots\tau^{j+1}(a)\tau^{j+1}(c) = d_{j+1}$$

for all $j \in \{0, \dots, n-2\}$. Let $b = d_0 + \cdots + d_{n-1}$. We claim that $b \neq 0$ for some c . Suppose this is not true, say $b = 0$ for all c . Then

$$0 = ac + (a\tau(a))\tau(c) + \cdots + (a\tau(a)\cdots\tau^{n-1}(a))\tau^{n-1}(c)$$

for every $c \in E$. This implies that $a = 0$ by Dedekind's theorem, a contradiction.

So let $c \in E$ be such that $b \neq 0$. Then

$$\begin{aligned} \tau(b) &= \tau(d_0) + \cdots + \tau(d_{n-1}) \\ &= \tau(ac) + \tau(a\tau(a)\tau(c)) + \cdots + \tau(\tau^{n-1}(c)) \\ &= \frac{1}{a}(d_1 + \cdots + d_{n-1}) + \tau^n(c) \\ &= \frac{1}{a}(d_0 + \cdots + d_{n-1}) \\ &= b/a. \end{aligned}$$

□

16.2. EXERCISE. Let E/K be a cyclic extension. Assume that $\text{Gal}(E/K)$ is generated by τ . Prove that for $a \in E$, $\text{trace}_{E/K}(a) = 0$ if and only if $a = b - \tau(b)$ for some $b \in E \setminus \{0\}$.

16.3. COROLLARY. Let $a, b, c \in \mathbb{Z}$ be such that $a^2 + b^2 = c^2$. Then

$$(a, b, c) = \begin{cases} \lambda(r^2 - s^2, 2rs, r^2 + s^2) & \text{or} \\ \lambda(2rs, r^2 - s^2, r^2 + s^2) \end{cases}$$

for some $r, s \in \mathbb{Z}$ and some $\lambda \in \mathbb{Z}$.

PROOF. We work with the extension $\mathbb{Q}(i)/\mathbb{Q}$. Note that $\text{Gal}(\mathbb{Q}(i), \mathbb{Q}) = \{\text{id}, \gamma\}$ is cyclic, where $\gamma: \mathbb{Q}(i) \rightarrow \mathbb{Q}(i)$, $z \mapsto \bar{z}$, is the complex conjugation. We may assume that $c \neq 0$, otherwise $a = b = 0$ and the result is trivial. Write $(a/c)^2 + (b/c)^2 = 1$ and let

$$\alpha = (a/c) + (b/c)i \in \mathbb{Q}(i).$$

Then $\text{norm}_{\mathbb{Q}(i)/\mathbb{Q}}(\alpha) = 1$. By Hilbert's theorem, there exists $\beta \in \mathbb{Q}(i) \setminus \{0\}$ such that

$$\alpha = a + bi = \frac{\beta}{\gamma(\beta)}.$$

Note that if $m \in \mathbb{Z} \setminus \{0\}$, then $\frac{m\beta}{\gamma(m\beta)} = \frac{\beta}{\gamma(\beta)}$. There exists $m \in \mathbb{Z} \setminus \{0\}$ such that $m\beta \in \mathbb{Z}[i]$, say $m\beta = r + is$ with $r, s \in \mathbb{Z}$ coprime. Then

$$\alpha = \frac{\beta}{\gamma(\beta)} = \frac{m\beta}{\gamma(m\beta)} = \frac{r + is}{r - is} = \frac{r^2 - s^2 + 2rsi}{r^2 + s^2} = \frac{r^2 - s^2}{r^2 + s^2} + \frac{2rs}{r^2 + s^2}i.$$

Note that

$$\gcd(r^2 - s^2, r^2 + s^2) = \gcd(r^2 - s^2, (r^2 - s^2) + (r^2 + s^2)) = \gcd(r^2 - s^2, 2r^2) \leq 2.$$

If the gcd is 1, then

$$(a, b, c) = \lambda(r^2 - s^2, 2rs, r^2 + s^2).$$

Otherwise

$$(a, b, c) = \lambda \left(\frac{r^2 - s^2}{2}, \frac{2rs}{2}, \frac{r^2 + s^2}{2} \right).$$

However, as r and s are coprime such that $r^2 - s^2 \equiv 0 \pmod{2}$, they must both be odd, and we can choose coprime integers u and w such that

$$\begin{aligned} r &= u + w, \\ s &= u - w. \end{aligned}$$

Substituting these in, we get

$$\frac{r^2 - s^2}{2} = 2uw, \quad \frac{2rs}{2} = u^2 - w^2, \quad \frac{r^2 + s^2}{2} = u^2 + w^2,$$

giving us the second triple in the statement. □

16.4. EXERCISE. Let $A, B \in \mathbb{Z}$ be such that $A^2 - 4B$ is not a square. Prove that a solution $(x, y, z) \in \mathbb{Z}^3$ of $x^2 + Axy + By^2 = z^2$ is proportional to

$$(r^2 - Bs^2, 2rs + As^2, r^2 + Ars + Bs^2).$$

16.5. PROPOSITION. Let $n \geq 2$ and K be a field containing a primitive n -root of one. If $a \in K^\times$ and E/K is a splitting field of $f = X^n - a$, then E/K is cyclic of degree d , where d divides n . Moreover,

$$d = \min\{k : a^k \in K^\times\},$$

where $K^\times = \{x \in K : x = y^n \text{ for some } y \in K\}$. Conversely, if E/K is cyclic of degree n , then E/K is a splitting field of an irreducible polynomial of the form $X^n - a$ for some $a \in K^\times$.

PROOF. A splitting field of f over K is of the form $K(\alpha)$, where $\alpha^n = a$. Thus $K(\alpha)/K$ is a Galois extension. If $\sigma \in \text{Gal}(K(\alpha)/K)$, then $\sigma(\alpha)$ is a root of f , so $\sigma(\alpha) = \omega_\sigma \alpha$, where $\omega_\sigma \in G_n(K)$. This means that there exists an injective map

$$\lambda: \text{Gal}(K(\alpha)/K) \rightarrow G_n(K), \quad \sigma \mapsto \omega_\sigma.$$

Moreover, λ is a group homomorphism, as

$$\sigma\tau(\alpha) = \sigma(\tau(\alpha)) = \sigma(\omega_\tau \alpha) = \omega_\tau \sigma(\alpha) = \omega_\tau \omega_\sigma \alpha.$$

Therefore $\text{Gal}(K(\alpha)/K)$ is isomorphic to a subgroup of $G_n(K)$. In particular, $\text{Gal}(K(\alpha)/K)$ is cyclic and $|\text{Gal}(K(\alpha)/K)|$ divides n .

Let $d = |\text{Gal}(K(\alpha)/K)|$. Since $a = \alpha^n$,

$$\text{norm}_{K(\alpha)/K}(\alpha)^n = \text{norm}_{K(\alpha)/K}(a) = a^d.$$

Thus $a^d \in K^\times$, as $\text{norm}_{K(\alpha)/K}(\alpha) \in K$. If $a^k \in K^\times$, say $a^k = c^n$ for some $c \in K$, then

$$c^n = a^k = (\alpha^n)^k = (\alpha^k)^n \implies \alpha^k = c\omega \in K$$

for some $\omega \in G_n(K)$. Thus α is a root of $X^k - \alpha^k \in K[X]$ and hence $k \geq d$.

Note that $f(\alpha, K) = X^d - \alpha^d$.

Let E/K be cyclic of degree n . Assume that $\text{Gal}(E/K) = \langle \sigma \rangle$. If ω is a primitive n -root of one,

$$\text{norm}_{E/K}(\omega) = \omega^n = 1.$$

By Hilbert's theorem 90, there exists a $b \in E^\times$ such that $\omega = b/\sigma(b)$. Thus $\sigma(b) = \omega^{-1}b$ and hence $\sigma^i(b) = \omega^{-i}b$ for all $i \geq 0$. Since $|\{b, \sigma(b), \dots, \sigma^{n-1}(b)\}| = n$, it follows that $E = K(b)$. Moreover,

$$\sigma(b^n) = \sigma(b)^n = (\omega^{-1}b)^n = b^n$$

and hence $b^n \in K$. This means that E/K is a decomposition field of $X^n - b^n$. Note that $X^n - b^n$ is irreducible, as $[E : K] = [K(b) : K] = n$. $\square$

16.6. PROPOSITION. Let K be a field of characteristic $p > 0$.

- 1) Let $a \in K$ and $f = X^p - X - a$. Then f is irreducible over K or all the roots of f belong to K . In the first case, if b is a root of f , then $K(b)/K$ is a cyclic extension of degree p .
- 2) Every cyclic extension of degree p is a splitting field of an irreducible polynomial of the form $X^p - X - a$.

PROOF. We first prove 1). Let K_0 be the prime field of K and C be an algebraic closure of K . Note that $K_0 \simeq \mathbb{Z}/p$. Let $b \in C$ be a root of f and let $x \in K_0$. Then

$$f(b+x) = (b+x)^p - (b+x) - a = (b^p - b - a) + (x^p - x) = 0$$

and thus $\{b+x : x \in K_0\} \subset C$ is the set of roots of f . Note that $f' = -1$, so f has no multiple roots.

§ 17. Symmetric polynomials

We claim that if $b \notin K$, then f is irreducible. If f is not irreducible, then $f = gh$ for some $g, h \in K[X]$ such that $0 < \deg g < p$. There exists a subset S of K_0 such that $g = \prod_{x \in S} (X - (b + x))$ in $C[X]$ and hence

$$|S|b + \sum_{x \in S} x = \sum_{x \in S} (b + x) \in K.$$

This implies that $|S|b \in K$ and hence, since $|S| \in K^\times$, it follows that $b \in K$.

Since $K(b)/K$ is a splitting field of a separable polynomial, $K(b)/K$ is a Galois extension. Moreover, $|\text{Gal}(K(b)/K)| = [K(b) : K] = p$ and hence $\text{Gal}(K(b)/K)$ is cyclic.

We now prove 2). Let E/K be cyclic of degree p . Assume that $\text{Gal}(E/K) = \langle \sigma \rangle$. Since $\text{trace}_{E/K}(1) = p = 0$, Hilbert's theorem implies that there exists $b \in E$ such that $\sigma(b) = b + 1$. In particular, $b \notin K$ and thus $E = K(b)$. Moreover, since

$$\sigma(b^p - b) = \sigma(b)^p - \sigma(b) = (b + 1)^p - (b + 1) = b^p - b,$$

it follows that $b^p - b \in K$. Thus $f(b, K) = X^p - X - (b^p - b) \in K[X]$. $\square$

§ 17. Symmetric polynomials

Let K be a field and $\{t_1, \dots, t_n\}$ be a commuting set of independent variables. Let $E = K(t_1, \dots, t_n)$ and $f = \prod_{i=1}^n (X - t_i) \in E[X]$. Then

$$f = X^n + \sum_{i=1}^n (-1)^i s_i X^{n-i},$$

where

$$\begin{aligned} s_1 &= t_1 + t_2 + \dots + t_n, \\ s_2 &= \sum_{1 \leq i < j \leq n} t_i t_j, \\ &\vdots \\ s_n &= t_1 t_2 \dots t_n. \end{aligned}$$

For example,

$$(X - t_1)(X - t_2)(X - t_3) = X^3 - (t_1 + t_2 + t_3)X^2 + (t_1 t_2 + t_2 t_3 + t_1 t_3)X - t_1 t_2 t_3.$$

The polynomials $s_1, s_2, \dots, s_n$ are known as the **elementary symmetric polynomials** in the variables $t_1, \dots, t_n$. Note that $\deg s_i = i$.

Let $\sigma \in \mathbb{S}_n$ and

$$\alpha_\sigma : K[t_1, \dots, t_n] \rightarrow K[t_1, \dots, t_n], \quad t_i \mapsto t_{\sigma(i)} \quad \text{for all } i.$$

Then α_σ is a bijective homomorphism of K -algebras. In fact, $\alpha_\sigma^{-1} = \alpha_{\sigma^{-1}}$. Note that

$$\alpha_\sigma(h(t_1, \dots, t_n)) = h(t_{\sigma(1)}, \dots, t_{\sigma(n)}).$$

Since α_σ is injective, it induces an element $\hat{\sigma} \in \text{Gal}(E/K)$ given by

$$\hat{\sigma} \left(\frac{h}{g} \right) = \frac{\alpha_\sigma(h)}{\alpha_\sigma(g)}.$$

The map $\mathbb{S}_n \rightarrow \text{Gal}(E/K)$, $\sigma \mapsto \hat{\sigma}$, is an injective group homomorphism. Thus

$$\{\hat{\sigma} : \sigma \in \mathbb{S}_n\} \simeq \mathbb{S}_n.$$

§ 17. Symmetric polynomials

17.1. DEFINITION. Let $g \in K[t_1, \dots, t_n]$. Then g is **symmetric** if $\widehat{\sigma}(g) = g$ for all $\sigma \in \mathbb{S}_n$.

We write P to denote the set of symmetric polynomials in $K[t_1, \dots, t_n]$. Clearly, P is a subalgebra of $K[t_1, \dots, t_n]$. The following statements hold:

- 1) $K \subseteq P$.
- 2) $\sum_{i=1}^n t_i^r \in P$ for all $r \geq 1$.
- 3) $s_i \in P$ for all i .
- 4) $K(P) \subseteq {}^G E$, where $G = \{\widehat{\sigma} : \sigma \in \mathbb{S}_n\}$.

Let $F = K(s_1, s_2, \dots, s_n)$. Then E/F is a Galois extension, as it is a splitting field of f .

17.2. PROPOSITION. $[E : F] \leq n!$.

PROOF. We proceed by induction on n . The case $n = 1$ is clear, as $E = F$. Assume that $n > 1$. Let $u_1, \dots, u_{n-1}$ be the elementary symmetric polynomials in $t_1, \dots, t_{n-1}$. Then

$$s_i = u_i + t_n u_{i-1}$$

for all $i \in \{1, \dots, n\}$, where $u_0 = 1$ and $u_n = 0$. Note that $u_1 = s_1 - t_n$ and $u_i = s_i - t_n u_{i-1}$ for all i . Since $K(s_1, \dots, s_n, t_n) = K(u_1, \dots, u_{n-1}, t_n)$,

$$F(t_n) = K(u_1, \dots, u_{n-1}, t_n) = K(t_n)(u_1, \dots, u_{n-1})$$

and

$$[E : F] = [E : F(t_n)][F(t_n) : F] \leq n[E : F(t_n)].$$

Note that $E = K(t_1, \dots, t_n) = K(t_n)(t_1, \dots, t_{n-1})$. By the inductive hypothesis,

$$[E : F(t_n)] \leq (n-1)!$$

and hence $[E : F] \leq n!$. □

17.3. THEOREM. ${}^G E = F$.

PROOF. By Artin's theorem,

$$[{}^G E : F] = \frac{[E : F]}{[E : {}^G E]} \leq \frac{n!}{[E : {}^G E]} = 1$$

and hence ${}^G E = F$. □

17.4. EXERCISE. Prove that $\text{Gal}(E/F) \simeq \mathbb{S}_n$.

17.5. EXERCISE. Prove that $\{s_1, \dots, s_n\}$ is algebraically independent over K .

17.6. EXERCISE. Prove that every symmetric polynomial in $t_1, \dots, t_n$ can be written as a rational fraction in $s_1, \dots, s_n$.

§ 18. Solvable groups

Let G be a group. If $x, y \in G$ we define the **commutator** of x and y as

$$[x, y] = xyx^{-1}y^{-1}.$$

Note that $[x, y] = 1$ if and only if $xy = yx$. Moreover, $[x, y]^{-1} = [y, x]$. The **commutator (or derived) subgroup** $[G, G]$ of G is defined as the subgroup of G generated by all commutators, i.e.

$$[G, G] = \langle [x, y] : x, y \in G \rangle.$$

This means that every element of $[G, G]$ is a finite product of commutators, so every element of $[G, G]$ is of the form $\prod_{i=1}^m [x_i, y_i]$. In general, the commutator subgroup is not equal to the set of commutators!

18.1. EXAMPLE. This example is taken from the book [2] of Carmichael. Let G be the subgroup of S_{16} generated by the permutations

$$\begin{aligned} a &= (13)(24), & b &= (57)(68), \\ c &= (911)(10\ 12), & d &= (13\ 15)(14\ 16), \\ e &= (13)(57)(9\ 11), & f &= (12)(34)(13\ 15), \\ g &= (56)(78)(13\ 14)(15\ 16), & h &= (9\ 10)(11\ 12). \end{aligned}$$

Then $[G, G]$ has order 16. However, the set $\{[x, y] : x, y \in G\}$ of commutators has 15 elements:

```
> S16 := Sym(16);
> a := S16 ! (1,3)(2,4);
> b := S16 ! (5,7)(6,8);
> c := S16 ! (9,11)(10,12);
> d := S16 ! (13,15)(14,16);
> e := S16 ! (1,3)(5,7)(9,11);
> f := S16 ! (1,2)(3,4)(13,15);
> g := S16 ! (5,6)(7,8)(13,14)(15,16);
> h := S16 ! (9,10)(11,12);
> G := PermutationGroup< 16 | a,b,c,d,e,f,g,h >;
> D := DerivedSubgroup(G);
> #D;
16
> #{ x*y*Inverse(x)*Inverse(y) : x in G, y in G };
15
> c*d in { x : x in D } \
> diff { u*v*Inverse(u)*Inverse(v) : u in G, v in G };
true
```


18.2. EXERCISE. Let G be a group. Prove the following facts:

- 1) G is abelian if and only if $[G, G] = \{1\}$.
- 2) $[G, G]$ is a normal subgroup of G .
- 3) $G/[G, G]$ is abelian.
- 4) If H is a subgroup of G and $[G, G] \subseteq H$, then H is normal in G .
- 5) If H is a normal subgroup of G , then G/H is abelian if and only if $[G, G] \subseteq H$.

18.3. DEFINITION. Let G be a group. The **derived series** of G is defined as $G^{(0)} = G$ and $G^{(k+1)} = [G^{(k)}, G^{(k)}]$ for $k \geq 0$.

18.4. EXERCISE. Prove that $G^{(k)}$ is normal in G for all k .

Why derived series? We cannot explain this here, but let us use the following notation. We write $G' = [G, G]$, $G'' = [G', G']$... Note that

$$G \supseteq G' \supseteq G'' \supseteq \dots$$

18.5. EXERCISE. Let $n \geq 3$. Prove that $[\mathbb{S}_n, \mathbb{S}_n] = \mathbb{A}_n$.

18.6. EXAMPLE. Let $K = \{\text{id}, (12)(34), (13)(24), (14)(23)\}$. Then K is a normal subgroup of $\mathbb{A}_4$. One proves that $[\mathbb{A}_4, \mathbb{A}_4] = K$.

A group G is said to be **simple** if there are no proper non-trivial normal subgroups of G . If p is a prime number, then the group $\mathbb{Z}/p$ of integers modulo p is a simple group. We will prove later that $\mathbb{A}_n$ is simple if $n \geq 5$.

18.7. EXAMPLE. Let $n \geq 5$. Since $\mathbb{A}_n$ is a non-abelian simple group, $[\mathbb{A}_n, \mathbb{A}_n] = \mathbb{A}_n$.

Let us show that $\mathbb{A}_5$ is a non-abelian simple group. Hence it is not solvable:

```
> G := Alt(5);
> IsAbelian(G);
false
> IsSimple(G);
true
> IsSolvable(G);
false
```

18.8. DEFINITION. A group G is **solvable** if and only if $G^{(m)} = \{1\}$ for some m .

Every abelian group is solvable.

18.9. EXERCISE. Prove that $\mathbb{S}_n$ is solvable if and only if $n \leq 4$.

Let us compute (with the computer software Oscar) the derived series of the symmetric group $\mathbb{S}_4$. The calculation shows that $\mathbb{S}_4$ is solvable:

```
> G := Sym(4);
> DerivedSeries(G);
[
  Symmetric group G acting on a set of cardinality 4
```

§ 18. Solvable groups

```

Order = 24 = 2^3 * 3
      (1, 2, 3, 4)
      (1, 2),
Permutation group acting on a set of cardinality 4
Order = 12 = 2^2 * 3
      (1, 2, 3)
      (2, 3, 4),
Permutation group acting on a set of cardinality 4
Order = 4 = 2^2
      (1, 4)(2, 3)
      (1, 3)(2, 4),
Permutation group acting on a set of cardinality 4
Order = 1
]
> IsSolvable(G);
true

```

18.10. PROPOSITION. *Let G be a group and H be a subgroup of G . The following statements hold:*

- 1) *If G is solvable, then H is solvable.*
- 2) *If H is normal in G and G is solvable, then G/H is solvable.*
- 3) *If H is normal in G and H and G/H are solvable, then G is solvable.*

PROOF. The first statement follows from the fact that $H^{(i)} \subseteq G^{(i)}$ holds for all i .

Assume now that H is normal in G . Let $Q = G/H$ and $\pi: G \rightarrow Q$ be the canonical map. By induction one proves that $\pi(G^{(i)}) = Q^{(i)}$ for all $i \geq 0$. The case where $i = 0$ is trivial, as π is surjective. If the result holds for some $i \geq 0$, then

$$\pi(G^{(i+1)}) = \pi([G^{(i)}, G^{(i)}]) = [\pi(G^{(i)}), \pi(G^{(i)})] = [Q^{(i)}, Q^{(i)}] = Q^{(i+1)}.$$

We now prove 2). Since G is solvable, $G^{(n)} = \{1\}$ for some n . Thus Q is solvable, as $Q^n = \pi(G^{(n)}) = \pi(\{1\}) = \{1\}$.

We finally prove 3). Since Q is solvable, $Q^{(n)} = \{1\}$ for some n . Moreover, since $\pi(G^{(n)}) = Q^{(n)} = \{1\}$, it follows that $G^{(n)} \subseteq H$. Since H is solvable,

$$G^{(n+m)} \subseteq (G^{(n)})^{(m)} \subseteq H^{(m)} = \{1\}$$

for some m . Thus G is solvable. □

An application:

18.11. PROPOSITION. *Let G be a finite p -group. Then G is solvable.*

PROOF. Assume the result is not true. Let G be a finite p -group of minimal order that is not solvable. Since G is a p -group, $Z(G) \neq \{1\}$. Since $|G|$ is minimal, $G/Z(G)$ is a solvable p -group. Since $Z(G)$ is abelian, $Z(G)$ is solvable. Now G is solvable by Proposition 18.10. □

Let G be a group. A proper subgroup N of G is said to be **maximal normal** if N is a normal subgroup of G and there is no other proper normal subgroup of G containing N .

18.12. EXERCISE. If a subgroup N of G is maximal (for the inclusion) and normal, then it is maximal normal. Show that the converse does not hold.

The following result is a direct consequence of the correspondence theorem:

18.13. EXERCISE. Let G be a group and N be a normal subgroup of G . Prove that N is maximal normal if and only if G/N is simple.

Maximal normal subgroups always exist in finite groups (they could be trivial). We can compute maximal normal subgroups as follows:

```
> MaximalNormalSubgroup(Sym(3));
Permutation group acting on a set of cardinality 3
Order = 3
      (1, 2, 3)
julia> maximal_normal_subgroups(quaternion_group(8))
3-element Vector{PcGroup}:
 Group([ y2, x ])
 Group([ y2, y ])
 Group([ y2, x*y ])
```

```
> MaximalNormalSubgroup(Alt(4));
Permutation group acting on a set of cardinality 4
Order = 4 = 2^2
      (1, 2)(3, 4)
      (1, 3)(2, 4)
```

18.14. EXERCISE. Let G be a finite solvable group. Prove that if G is simple, then G is cyclic of prime order.

The following result will be important later:

18.15. PROPOSITION. *Every finite solvable group contains a normal subgroup of prime index.*

PROOF. Let G be a finite solvable group. Let M be a maximal normal subgroup of G (there is at least one, as G is finite). Since G/M is simple and solvable (see Proposition 18.10), G/M is cyclic of prime order by Exercise 18.14. $\square$

We finish this discussion with two important theorems (without proof) about finite solvable groups.

18.16. THEOREM (Burnside). *Let p and q be prime numbers. If G is a group of order $p^a q^b$, then G is solvable.*

The proof appears in courses on the representation theory of finite groups.

18.17. THEOREM (Feit–Thompson). *Every finite group of odd order is solvable.*

The proof of the theorem is extremely hard. It occupies a full volume of **Pacific Journal of Mathematics**, see [3].

§ 19. Simplicity of the alternating simple group

We will present a family of non-abelian simple groups. We start with some exercises.

19.1. EXERCISE. Let G be a group. Prove that G is simple if and only if $\{(g, g) : g \in G\}$ is a maximal subgroup of $G \times G$.

19.2. EXERCISE. Prove that $\mathbb{A}_n$ is generated by 3-cycles.

19.3. EXERCISE. Compute the commutator subgroup of $\mathbb{A}_n$ for $n \geq 2$.

Note that $\mathbb{A}_2$ and $\mathbb{A}_3$ are abelian. For $\mathbb{A}_4$, one proves that

$$[\mathbb{A}_4, \mathbb{A}_4] = \{\text{id}, (12)(34), (13)(24), (14)(23)\}.$$

Finally, $[\mathbb{A}_n, \mathbb{A}_n] = \mathbb{A}_n$ for $n \geq 5$.

Let us compute some commutator subgroups (and the inclusion group homomorphism) with the computer:

```
> S3 := Sym(3);
> DerivedSubgroup(S3);
Permutation group acting on a set of cardinality 3
Order = 3
(1, 2, 3)
```

19.4. EXERCISE. Let $n \geq 3$. Prove that $[\mathbb{S}_n, \mathbb{S}_n] = \mathbb{A}_n$.

Recall that every normal subgroup is a union of conjugacy classes. The group $\mathbb{A}_5$ has conjugacy classes of sizes 1, 15, 20, 12 and 12. It follows that the only possible normal subgroups of $\mathbb{A}_5$ are $\{\text{id}\}$ and $\mathbb{A}_5$.

```
> A5 := Alt(5);
> ConjugacyClasses(A5);
Conjugacy Classes of group A5
-----
[1]      Order 1      Length 1
      Rep Id(A5)

[2]      Order 2      Length 15
      Rep (1, 2)(3, 4)

[3]      Order 3      Length 20
      Rep (1, 2, 3)

[4]      Order 5      Length 12
      Rep (1, 2, 3, 4, 5)

[5]      Order 5      Length 12
      Rep (1, 3, 4, 5, 2)
```

§ 19. Simplicity of the alternating simple group

19.5. THEOREM (Jordan). *Let $n \geq 5$. Then $\mathbb{A}_n$ is simple.*

Before proving the theorem, we need some preliminary results.

Every permutation $\rho \in \mathbb{S}_n$ decomposes as a product of disjoint cycles, say

$$\rho = (a_1 \cdots a_r)(b_1 \cdots b_s) \cdots (c_1 \cdots c_t)$$

where, by convention, we do not write cycles of length one. The cyclic structure of ρ is, by definition, the (unordered) sequence of integers $r, s, \dots, t$, where, again by convention, we omit fixed points. For example, the cyclic structure of the transposition (ab) is 2, of $(abc)(d)$ is 3 and of $(123)(45)(789a)(bcd)(d)$ is 2, 3, 3, 4.

19.6. LEMMA. *If both permutations $\rho_1 \in \mathbb{S}_n$ and $\rho_2 \in \mathbb{S}_n$ have the same cyclic structure, then $\rho_2 = \sigma \rho_1 \sigma^{-1}$ for some $\sigma \in \mathbb{S}_n$.*

PROOF. Assume that

$$\begin{aligned}\rho_1 &= (a_1 \cdots a_r)(b_1 \cdots b_s) \cdots (c_1 \cdots c_t), \\ \rho_2 &= (x_1 \cdots x_r)(y_1 \cdots y_s) \cdots (z_1 \cdots z_t).\end{aligned}$$

Let

$$\text{Fix}(\rho_1) = \{x \in \{1, \dots, n\} : \rho_1(x) = x\} = \{k_1, \dots, k_m\}, \quad \text{Fix}(\rho_2) = \{l_1, \dots, l_m\}$$

be the fixed points of the permutations ρ_1 and ρ_2 , respectively. Then

$$\sigma(x) = \begin{cases} x_j & \text{if } x = a_j \text{ for some } j, \\ y_j & \text{if } x = b_j \text{ for some } j, \\ \vdots & \\ z_j & \text{if } x = c_j \text{ for some } j, \\ l_j & \text{if } x = k_j \text{ for some } j, \end{cases}$$

is such that $\sigma \rho_1 \sigma^{-1} = \rho_2$. □

What happens with the alternating group?

19.7. LEMMA. *If $\rho_1, \rho_2 \in \mathbb{S}_n$ are conjugate in $\mathbb{S}_n$ and $|\text{Fix}(\rho_1)| \geq 2$, then $\mu \rho_1 \mu^{-1} = \rho_2$ for some $\mu \in \mathbb{A}_n$.*

PROOF. Assume that $\rho_2 = \sigma \rho_1 \sigma^{-1}$ for some $\sigma \in \mathbb{S}_n$. There are $a, b \in \{1, \dots, n\}$ such that $\rho_1(a) = a$, $\rho_1(b) = b$ and $a \neq b$. Let

$$\mu = \begin{cases} \sigma & \text{if } \sigma \in \mathbb{A}_n, \\ \sigma(ab) & \text{otherwise.} \end{cases}$$

Then $\mu \in \mathbb{A}_n$ and $\mu \rho_1 \mu^{-1} = \rho_2$, as (ab) commutes with ρ_1 . □

Let us discuss some examples.

19.8. EXAMPLE. If $\rho_1 = (23)(156)$ and $\rho_2 = (45)(123)$, then $\rho_2 = \sigma \rho_1 \sigma^{-1}$ for

$$\sigma = \begin{pmatrix} 123456 \\ 145623 \end{pmatrix}.$$

19.9. EXAMPLE. The permutations $\rho_1 = (123) \in \mathbb{S}_3$ and $\rho_2 = (132) \in \mathbb{S}_3$ are conjugate in $\mathbb{S}_3$, as $(123) = \sigma(132)\sigma^{-1}$ if $\sigma = (23)$. However, ρ_1 and ρ_2 are not conjugate in $\mathbb{A}_3$.

§ 20. Radical extensions

Now we are ready to prove the theorem.

PROOF OF THEOREM 19.5. Let $N \neq \{\text{id}\}$ be a normal subgroup of $\mathbb{A}_n$. If $(abc) \in N$, then every 3-cycle belongs to N , because all 3-cycles are conjugate in $\mathbb{S}_n$, and the previous lemma states that $(ijk) = \mu(abc)\mu^{-1} \in N$ for some $\mu \in \mathbb{A}_n$. Thus $N = \mathbb{A}_n$.

We claim that N contains a 3-cycle. Since $N \neq \{\text{id}\}$, there exists $\sigma \in N \setminus \{\text{id}\}$. Let $m = |\sigma|$ and let p be a prime number dividing m . Then $\tau = \sigma^{m/p}$ has order p and hence $\tau = \rho_1 \cdots \rho_s$, where the ρ_j 's are disjoint p -cycles.

If $p = 2$, then $1 = \text{sign}(\tau) = (-1)^s$. Thus s is even. Write

$$\tau = (ab)(cd)\rho_3 \cdots \rho_s.$$

Since $\rho_3 \cdots \rho_s$ commutes with (abc) and (acb) ,

$$\underbrace{(abc)\tau(abc)^{-1}\tau^{-1}}_{\in N} = (abc)(ab)(cd)(acb)(ab)(cd) = (ac)(bd).$$

Hence $(ac)(bd) \in N$. Let $e \in \{1, \dots, n\} \setminus \{a, b, c, d\}$. Then

$$(ae)(bd) = (aec) \underbrace{(ac)(bd)}_{\in N} (aec)^{-1} \in N$$

and therefore

$$(aec) = (ac)(ae) = (ac)(bd)(ae)(bd) \in N.$$

If $p = 3$, without loss of generality, we may assume that $s \geq 2$ (otherwise, τ would be a 3-cycle). Then $\tau = (abc)(def)\rho_3 \cdots \rho_s$. Since (bcd) commutes with $\rho_3 \cdots \rho_s$ and N is normal in $\mathbb{A}_n$,

$$\underbrace{(bcd)\tau(bcd)^{-1}\tau^{-1}}_{\in N} = (bcd)(abc)(def)(bdc)(acb)(dfe) = (adbce)$$

and therefore

$$(adc) = (adb)(adbce)(adb)^{-1}(adbce)^{-1} \in N.$$

If $p > 3$, then $\tau = (abcd \cdots z)\rho_2 \cdots \rho_s$. In particular, (abc) commutes with $\rho_2 \cdots \rho_s$. Then

$$(abd) = (abc)\tau(abc)^{-1}\tau^{-1} \in N. \quad \square$$

As an application, we compute the normal subgroups of the symmetric group $\mathbb{S}_n$.

19.10. EXERCISE. Compute the list of normal subgroups of $\mathbb{S}_n$ for $n \geq 2$.

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20.1. DEFINITION. An extension E/K is said to be **pure** of type m if $E = K(x)$ for some x such that $x^m \in K$.

Note that if $E = K(x)$ is a pure extension of type m and K contains all m -th roots of one, then E/K is a splitting field of $X^m - x^m$.

20.2. DEFINITION. The sequence $K = R_0 \subseteq R_1 \subseteq \cdots \subseteq R_m$ of fields is said to be a **radical tower** if each R_{i+1}/R_i is pure. In this case, R_m/K is a **radical extension**.

Note that radical extensions are finite.

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20.3. EXAMPLE. Let E be a splitting field of $X^4 - 2$ over $\mathbb{Q}$. Then $E/\mathbb{Q}$ is radical, as $E = \mathbb{Q}(\sqrt[4]{2}, i)$.

20.4. EXAMPLE. Let $\alpha, \beta \in \mathbb{C}$ be such that $\alpha^2 = 2$ and $\beta^5 = 1 + \alpha$. The number $\sqrt[5]{1 + \sqrt{2}}$ belongs to the radical extension $\mathbb{Q}(\alpha, \beta)/\mathbb{Q}$.

20.5. THEOREM. *Let K be of characteristic zero and R/K be a radical extension. If E/K is a subextension of R/K , then $\text{Gal}(E/K)$ is solvable.*

PROOF. Without loss of generality, we may assume that E/K is a Galois extension. To prove this fact, let $G = \text{Gal}(E/K)$ and $F = {}^G E$. Then E/F is a Galois extension and $\text{Gal}(E/F) = G$ by Artin's theorem. Thus, replacing K by F if needed, we may assume that E/K is Galois.

Let L be the normal closure of R in some algebraic closure C that contains R . Note that if $R = K(x_1, \dots, x_m)$, then

$$L = K(\{\sigma_i(x_j) : 1 \leq i \leq s, 1 \leq j \leq m\}),$$

where $\text{Hom}(R/K, C/K) = \{\sigma_1, \dots, \sigma_s\}$.

CLAIM. L/K is radical.

Since $x_j^{a_j} \in K(x_1, \dots, x_{j-1})$ for some integer a_j ,

$$\sigma_i(x_j)^{a_j} = \sigma_i(x_j^{a_j}) \in \sigma_i(K(x_1, \dots, x_{j-1})) = K(\sigma_i(x_1), \dots, \sigma_i(x_{j-1}))$$

Thus L/K is radical and Galois.

We may assume then that E/K and R/K are both Galois.

Since $\text{Gal}(E/K) \simeq \text{Gal}(R/K)/\text{Gal}(R/E)$, we only need to prove that $\text{Gal}(R/K)$ is solvable.

For a positive integer n , let ξ be a primitive n -th root of one (in some algebraic closure of K that contains R). Consider the diagram

$$\begin{array}{ccc} & R(\xi) & \\ & \swarrow \quad \searrow & \\ R & & K(\xi) \\ & \swarrow \quad \searrow & \\ & K & \end{array}$$

Then

- 1) $K(\xi)/K$ and $R(\xi)/R$ are abelian.
- 2) $R(\xi)/K$ is Galois.
- 3) $\text{Gal}(R/K) \simeq \text{Gal}(R(\xi)/K)/\text{Gal}(R(\xi)/R)$.
- 4) $\text{Gal}(K(\xi)/K) \simeq \text{Gal}(R(\xi)/K)/\text{Gal}(R(\xi)/K(\xi))$.

The third item implies that we need to show that $\text{Gal}(R(\xi)/K)$ is solvable. By the fourth item, it suffices to show that $\text{Gal}(R(\xi)/K(\xi))$ is solvable (because $\text{Gal}(K(\xi)/K)$ is abelian and hence solvable).

Since $R = K(x_1, \dots, x_m)$,

$$R(\xi) = K(x_1, \dots, x_m, \xi) = K(\xi)(x_1, \dots, x_m)$$

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and hence $R(\xi)/K(\xi)$ is radical. This means that without loss of generality, we may assume that K contains primitive n -roots of one. For example, if $R = K(x_1, \dots, x_m)$ and $x_i^{a_i} \in K(x_1, \dots, x_{i-1})$, then we may assume that K contains a primitive a_i -root of one. We proceed by induction on m . The case $m = 0$ is trivial. Assume that the claim holds for some $m \geq 0$. Let $L = K(x_1)$. Then L/K is a splitting field of $X^{a_1} - x_1^{a_1}$, and hence L/K is a cyclic extension. Thus $\text{Gal}(L/K)$ is cyclic (and hence, in particular, solvable). Let H be the subgroup that corresponds to L , that is $H = \text{Gal}(R/L)$ (here, we use Galois' correspondence). Then H is normal in $\text{Gal}(R/K)$. Since $R = K(x_1, \dots, x_m) = L(x_2, \dots, x_m)$, R/L is radical and Galois. By the inductive hypothesis, $\text{Gal}(R/L)$ is solvable. Since

$$\text{Gal}(L/K) \simeq \text{Gal}(R/K) / \text{Gal}(R/L),$$

it follows that $\text{Gal}(R/K)$ is solvable. $\square$

20.6. DEFINITION. Let $f \in K[X]$ and E be a splitting field of f over K . We say that f is **solvable by radicals** if there is a radical extension R/K such that $E \subseteq R$.

The general polynomial of degree two is solvable by radicals, as its Galois group is solvable (in fact, isomorphic to $\mathbb{S}_2$).

20.7. EXERCISE. Prove that $f = X^2 - s_1X + s_2 \in \mathbb{Q}[X]$ is solvable by radicals.

Theorem 20.5 translates into the following result:

20.8. EXERCISE. Let K be a field of characteristic zero. If $f \in K[X]$ is solvable by radicals, then $\text{Gal}(f, K)$ is solvable.

As a consequence, the general polynomial of degree $n \geq 5$ is not solvable by radicals, as its Galois group contains a subgroup isomorphic to $\mathbb{S}_5$.

20.9. EXAMPLE. Let p be a prime number and $f = X^5 - 2pX + p \in \mathbb{Q}[X]$. We claim that f is not solvable by radicals.

Note that $f' = 5X^4 - 2p$. Then $\alpha = \sqrt[4]{2p/5}$ and $\beta = -\sqrt[4]{2p/5}$ are critical points. Since $f(\alpha) < 0$ and $f(\beta) > 0$, it follows that f has exactly three real roots. Let $x_1, x_2 \in \mathbb{C} \setminus \mathbb{R}$ and $x_3, x_4, x_5 \in \mathbb{R}$ be the roots of f .

By Eisenstein's theorem, f is irreducible.

Let $E/\mathbb{Q}$ be a splitting field of f . Then $\text{Gal}(f, \mathbb{Q}) = \text{Gal}(E/\mathbb{Q})$ is isomorphic to a subgroup G of $\mathbb{S}_5$. Since f is irreducible, 5 divides $[E : \mathbb{Q}] = |G|$. In particular, by Cauchy's theorem, G contains an element σ of order five. This element is a 5-cycle, so without loss of generality, we may assume that $\sigma = (x_1x_2x_3x_4x_5)$. Note that $(x_1x_2) \in G$. Thus $G \simeq \mathbb{S}_5$ and hence G is not solvable.

20.10. EXERCISE. Let $f = X^6 + 2X^5 - 5X^4 + 9X^3 - 5X^2 + 2X + 1 \in \mathbb{Q}[X]$. Prove that f is solvable by radicals.

It is now time to prove Galois' great theorem on solvability of polynomials.

20.11. THEOREM (Galois). *Let K be a field of characteristic zero and $f \in K[X]$. Then f is solvable by radicals if and only if $\text{Gal}(f, K)$ is solvable.*

We proved in Theorem 20.5 that solvable polynomials have solvable Galois groups. For the converse, we need two auxiliary results.

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20.12. LEMMA. *Let E/K be a Galois extension of prime degree p . Assume that K admits a primitive p -root of one. Then $E = K(\beta)$ where $\beta^p \in K$.*

PROOF. Assume that $\text{Gal}(E/K) = \langle \sigma \rangle$. Let $\omega \in K$ be a primitive p -root of one. Then $\text{norm}_{E/K}(\omega) = \omega^p = 1$. By Hilbert's theorem, $\omega = \beta/\sigma(\beta)$ for some $\beta \in E$. Note that $\beta \notin K$, as $\omega \neq 1$. Moreover,

$$\sigma(\beta^p) = (\beta\omega^{-1})^p = \beta^p \in {}^{\text{Gal}(E/K)}E = K.$$

Since $K \subseteq K(\beta) \subseteq E$ and $[E : K] = p$, we conclude that $E = K(\beta)$ with $\beta^p \in K$. $\square$

20.13. EXERCISE. Let E/K be a splitting field of $f \in K[X]$ and K^*/K be an extension. If E^*/K^* is a splitting field of f containing E , then

$$\text{Gal}(E^*/K^*) \rightarrow \text{Gal}(E/K), \quad \sigma \mapsto \sigma|_E,$$

is an injective group homomorphism.

Now Theorem 20.11 will follow from the following theorem.

20.14. THEOREM. *Let K be a field of characteristic zero and E/K be a finite Galois extension. If $\text{Gal}(E/K)$ is solvable, then E can be embedded in a radical extension.*

PROOF. Let $G = \text{Gal}(E/K)$. Since G is solvable, by Proposition 18.15, there exists a normal subgroup H of G of prime index p . Let ω be a primitive p -th root of one. (It exists because K is a field of characteristic zero.)

We proceed by induction on $[E : K]$.

If $[E : K] = 1$, there is nothing to prove. So assume that $[E : K] > 1$.

We first assume that $\omega \in K$. The group $\text{Gal}(E/{}^HE)$ is solvable, as it is a subgroup of G . Moreover, since

$$[E : {}^HE] < [E : K],$$

the inductive hypothesis implies that $E/{}^HE$ can be embedded in a radical extension, so there exists a radical tower

$$(20.1) \quad {}^HE \subseteq R_1 \subseteq R_2 \subseteq \cdots \subseteq R_m,$$

where $E \subseteq R_m$. Now ${}^HE/K$ is a Galois extension, as H is normal in G . Moreover,

$$[{}^HE : K] = (G : H) = p.$$

Since $\omega \in K$, Lemma 20.12 implies that ${}^HE = K(\beta)$ for some β such that $\beta^p \in K$. The radical tower 20.1 can be extended by adding $K \subseteq {}^HE$.

For the general case, let $K^* = K(\omega)$ and $E^* = E(\omega)$. Then E^*/K^* is a Galois extension with Galois group $\text{Gal}(E^*/K^*)$. By Exercise 20.13, $\text{Gal}(E^*/K^*)$ is solvable. By the previous part, E^* and E can be embedded in a radical extension R^*/K^* , so there exists a radical tower

$$(20.2) \quad K^* \subseteq R_1^* \subseteq R_2^* \subseteq \cdots \subseteq R_n^*.$$

Since $K^* = K(\omega)$ is a pure extension, the radical tower (20.2) can be extended by adding $K \subseteq K^*$. $\square$

§ 21. Group cohomology

Let G be a group and A be a **(left) G -module**. This means that A is an abelian group together with a map

$$G \times A \rightarrow A, \quad (g, a) \mapsto g \cdot a$$

such that $1 \cdot a = a$ for all $a \in A$, $(gh) \cdot a = g \cdot (h \cdot a)$ for all $g, h \in G$ and $a \in A$ and $g \cdot (a + b) = g \cdot a + g \cdot b$ for all $g \in G$ and $a, b \in A$.

21.1. EXAMPLE. The group $\text{Gal}(\mathbb{C}/\mathbb{R})$ acts on $\mathbb{C}$ and $\mathbb{C}^\times$. Moreover, it acts trivially on $\mathbb{R}$ and $\mathbb{R}^\times$.

More generally, if E/K is a finite Galois extension, then the Galois group $\text{Gal}(E/K)$ acts on E and $E^\times$.

21.2. DEFINITION. Let G be a group and M and N be G -modules. A map $f: M \rightarrow N$ is a **homomorphism** of G -modules if $f(\sigma \cdot m) = \sigma \cdot f(m)$ for all $m \in M$ and $\sigma \in G$.

21.3. DEFINITION. Let G be a group and M be a G -module. The submodule of **G -invariants** is defined as

$$M^G = \{m \in M : \sigma \cdot m = m \text{ for all } \sigma \in G\}.$$

Note that M^G is the largest submodule of the G -module M where G acts trivially. For example, if $G = \text{Gal}(E/K)$, then $E^G = K$.

21.4. PROPOSITION. *Let G be a group. If the sequence of G -modules and G -module homomorphism*

$$0 \longrightarrow P \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$$

is exact, then

$$0 \longrightarrow P^G \xrightarrow{\alpha^0} M^G \xrightarrow{\beta^0} N^G$$

is exact, where α^0 is the restriction $\alpha|_{P^G}$ of α to P^G and β^0 is the restriction $\beta|_{M^G}$ of β to M^G .

PROOF. Since α is injective, the restriction α^0 is injective.

Note that $\ker \beta^0 = \ker \beta \cap M^G \subseteq \ker \beta$.

We claim that $\alpha^0(P^G) = \alpha(P) \cap M^G$. If $m \in \alpha(P) \cap M^G$, then $\alpha(p) = m$ for some $p \in P$ and $\sigma \cdot m = m$. Since

$$\alpha(p) = m = \sigma \cdot m = \sigma \cdot \alpha(p) = \alpha(\sigma \cdot p),$$

$\sigma \cdot p - p \in \ker \alpha = \{0\}$. Hence $\sigma \cdot p = p$ and $p \in P^G$. Conversely, if $m \in \alpha^0(P^G)$, then $m = \alpha(p)$ for some $p \in P^G$. If $\sigma \in G$, then

$$\sigma \cdot m = \sigma \cdot \alpha(p) = \alpha(\sigma \cdot p) = \alpha(p) = m.$$

Hence $m \in M^G \cap \alpha(P)$.

Now

$$\alpha^0(P^G) = \alpha(P) \cap M^G = \ker \beta \cap M^G = \ker \beta^0. \quad \square$$

Note that in the previous proposition, we did not prove that the map $\beta|_{M^G}$ is surjective.

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21.5. EXAMPLE. Let $G = \text{Gal}(\mathbb{C}/\mathbb{R})$. Consider the following exact sequence of G -modules:

$$1 \longrightarrow \{-1, 1\} \longrightarrow \mathbb{C}^\times \xrightarrow{\beta} \mathbb{C}^\times \longrightarrow 1$$

where $\beta(z) = z^2$. Note that β is surjective. Take invariants to obtain the sequence

$$0 \longrightarrow \{-1, 1\} \longrightarrow \mathbb{R}^\times \xrightarrow{\beta^0} \mathbb{R}^\times$$

where $\beta^0(x) = x^2$. Note that β^0 is not surjective!

21.6. DEFINITION. Let G be a group and M be a G -module. We define

$$H^0(G, M) = M^G,$$

$$C^1(G, M) = \{\phi: G \rightarrow M : \phi \text{ is a map}\},$$

$$Z^1(G, M) = \{\phi \in C^1(G, M) : \phi(\sigma\tau) = \phi(\sigma) + \sigma \cdot \phi(\tau) \text{ for all } \sigma, \tau \in G\},$$

Note that $Z^1(G, M)$ is an abelian group with the operation

$$(\phi + \phi_1)(\sigma) = \phi(\sigma) + \phi_1(\sigma).$$

Moreover, if $\phi \in Z^1(G, M)$, then $\phi(1_G) = 0_M$. To prove this fact, note that

$$\phi(1_G) = \phi(1_G 1_G) = \phi(1_G) + 1_G \cdot \phi(1_G) = \phi(1_G) + \phi(1_G)$$

implies that $\phi(1_G) = 0_M$.

21.7. EXAMPLE. Let G be a group and M be a G -module. Fix $m \in M$. Then the map $\phi: G \rightarrow M$, $\phi(\sigma) = \sigma \cdot m - m$, is an element of $Z^1(G, M)$, because

$$\begin{aligned} \phi(\sigma\tau) &= (\sigma\tau) \cdot m - m \\ &= (\sigma\tau) \cdot m - \sigma \cdot m + \sigma \cdot m - m \\ &= \sigma \cdot (\tau \cdot m - m) + \sigma \cdot m - m \\ &= \sigma \cdot \phi(\tau) + \phi(\sigma) \end{aligned}$$

for all $\sigma, \tau \in G$.

21.8. DEFINITION. Let G be a group and M be a G -module. The set $B^1(G, M)$ of **coboundaries** is the set of elements $\phi \in C^1(G, M)$ such that there is a fixed $m \in M$ such that $\phi(\sigma) = \sigma \cdot m - m$ for all $\sigma \in G$.

We proved in Example 21.7 that $B^1(G, M) \subseteq Z^1(G, M)$. A direct calculation shows that, in fact, $B^1(G, M)$ is a subgroup of $Z^1(G, M)$.

21.9. DEFINITION. Let G be a group and M be a G -module. The **first cohomology group** of G with coefficients in M is defined as the quotient

$$H^1(G, M) = Z^1(G, M) / B^1(G, M).$$

21.10. EXAMPLE. If G acts trivially on M , then

$$H^0(G, M) = M^G = M, \quad B^1(G, M) = \{0\}, \quad Z^1(G, M) = \text{Hom}(G, M).$$

Hence $H^1(G, M) \simeq \text{Hom}(G, M)$.

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21.11. EXAMPLE. Let $G = \text{Gal}(\mathbb{C}/\mathbb{R}) = \{\text{id}, \gamma\}$, where $\gamma: \mathbb{C} \rightarrow \mathbb{C}$, $z \mapsto \bar{z}$, is the complex conjugation. Then

$$H^0(G, \mathbb{R}^\times) = (\mathbb{R}^\times)^G = \mathbb{R}^\times.$$

Since G acts trivially on $\mathbb{R}^\times$,

$$H^1(G, \mathbb{R}^\times) = \text{Hom}(G, \mathbb{R}^\times) \simeq \text{Hom}(G, \{-1, 1\}) \simeq \mathbb{Z}/2.$$

The following lemma will be useful.

21.12. LEMMA. Let G be a group and $\alpha: M \rightarrow N$ be a homomorphism of G -modules. Then

$$\alpha^1: H^1(G, M) \rightarrow H^1(G, N), \quad \phi + B^1(G, M) \mapsto \alpha \circ \phi + B^1(G, N),$$

is a group homomorphism.

PROOF. Let us prove that the map α^1 is well-defined. If $\phi - \phi' \in B^1(G, M)$, then there exists a fixed $m \in M$ such that $(\phi - \phi')(\sigma) = \sigma \cdot m - m$ for all $\sigma \in G$. Let $n = \alpha(m) \in N$. For $\sigma \in G$,

$$\alpha((\phi - \phi')(\sigma)) = \alpha(\sigma \cdot m - m) = \sigma \cdot \alpha(m) - \alpha(m) = \sigma \cdot n - n.$$

Thus $\alpha \circ \phi - \alpha \circ \phi' \in B^1(G, N)$.

We now prove that α^1 is a group homomorphism. If $\phi, \phi' \in Z^1(G, M)$, then

$$\begin{aligned} \alpha^1(\phi + B^1(G, M) + \phi' + B^1(G, M)) &= \alpha^1(\phi + \phi' + B^1(G, M)) \\ &= \alpha \circ (\phi + \phi') + B^1(G, N) \\ &= \alpha \circ \phi + \alpha \circ \phi' + B^1(G, N) \\ &= \alpha \circ \phi + B^1(G, N) + \alpha \circ \phi' + B^1(G, N) \\ &= \alpha^1(\phi + B^1(G, M)) + \alpha^1(\phi' + B^1(G, M)). \quad \square \end{aligned}$$

We will provide a detailed proof of the upcoming result. The theorem will be established by applying a **diagram chasing** technique, a widely used tool in homological algebra. The proof is tedious and may seem intricate, but the method essentially involves “chasing” elements around a (commutative) diagram until we attain the desired result.

21.13. THEOREM. Let G be a group and

$$0 \longrightarrow P \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$$

be an exact sequence of G -modules and G -module homomorphism. Then there exists a **connection homomorphism** δ such that the sequence

$$(21.1) \quad \begin{array}{ccccccc} 0 & \longrightarrow & H^0(G, P) & \xrightarrow{\alpha^0} & H^0(G, M) & \xrightarrow{\beta^0} & H^0(G, N) \\ & & & & \searrow \delta & & \\ & & H^1(G, P) & \xrightarrow{\alpha^1} & H^1(G, M) & \xrightarrow{\beta^1} & H^1(G, N) \end{array}$$

of abelian groups and group homomorphisms is exact.

PROOF. By Proposition 21.4, the long sequence (21.1) is exact at $H^0(G, P) = P^G$, $H^0(G, M) = M^G$ and $H^0(G, N) = N^G$. Note that, in particular, $\alpha: P \rightarrow \alpha(P)$ is a bijective group homomorphism.

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Let us construct the connecting homomorphism $\delta: H^0(G, N) \rightarrow H^1(G, P)$. For $n \in N^G$, let $m \in M$ be such that $\beta(m) = n$. We define $\delta(n) = \phi + B^1(G, P)$, where

$$\phi(\sigma) = \alpha^{-1}(\sigma \cdot m - m).$$

Note that $\sigma \cdot m - m \in \text{im } \alpha = \ker \beta$, as

$$\beta(\sigma \cdot m - m) = \sigma \cdot \beta(m) - \beta(m) = \sigma \cdot n - n = 0.$$

Let us prove that the map δ is well-defined: if $m, m' \in M$ are such that

$$\beta(m) = \beta(m') = n,$$

then $m - m' \in \ker \beta = \alpha(P)$. For $\sigma \in G$, write $\phi'(\sigma) = \sigma \cdot m' - m'$. Since $m - m' = \alpha(p)$ for some $p \in P$ and α^{-1} is a homomorphism of G -modules,

$$\begin{aligned} \phi(\sigma) - \phi'(\sigma) &= \alpha^{-1}(\sigma \cdot m - m) - \alpha^{-1}(\sigma \cdot m' - m') \\ &= \alpha^{-1}(\sigma \cdot (m - m')) - \alpha^{-1}(m - m') \\ &= \alpha^{-1}(\sigma \cdot \alpha(p) - \alpha(p)) \\ &= \sigma \cdot p - p. \end{aligned}$$

Thus $\phi - \phi' \in B^1(G, P)$.

Note that $\phi \in Z^1(G, P)$, because

$$\begin{aligned} \phi(\sigma\tau) &= \alpha^{-1}((\sigma\tau) \cdot m - m) \\ &= \alpha^{-1}((\sigma\tau) \cdot m - \sigma \cdot m + \sigma \cdot m - m) \\ &= \alpha^{-1}(\sigma \cdot (\tau \cdot m - m)) + \alpha^{-1}(\sigma \cdot m - m) \\ &= \sigma \cdot \phi(\tau) + \phi(\sigma) \end{aligned}$$

holds for all $\sigma, \tau \in G$.

We now prove that the sequence (21.1) is exact at $H^0(G, N) = N^G$. We need to prove that $\ker \delta = \text{im } \beta^0$. To prove $\supseteq$ note that if $m \in M^G$ is such that $\delta(\beta(m)) = \phi + B^1(G, P)$, then

$$\phi(\sigma) = \alpha^{-1}(\sigma \cdot m - m) = 0.$$

Conversely, if $n \in \ker \delta$, then there exists $m \in M$ such that $\beta(m) = n$ and

$$\delta(\beta(m)) = \phi + B^1(G, P),$$

where $\phi \in B^1(G, P)$ and $\phi(\sigma) = \alpha^{-1}(\sigma \cdot m - m)$ for all $\sigma \in G$. Since $\phi \in B^1(G, P)$, there exists $p \in P$ such that $\phi(\sigma) = \sigma \cdot p - p$ for all $\sigma \in G$. Note that

$$\beta(m - \alpha(p)) = \beta(m) - \beta(\alpha(p)) = \beta(m) = n.$$

Moreover, $m - \alpha(p) \in M^G$, as $\sigma \cdot (m - \alpha(p)) = m - \alpha(p)$. Hence $n \in \text{im } \beta^0$.

We now prove that (21.1) is exact at $H^1(G, P)$, that is $\text{im } \delta = \ker \alpha^1$. To prove $\subseteq$ note that for $n \in N^G$, $\delta(n) = \phi + B^1(G, P)$, where $\phi(\sigma) = \alpha^{-1}(\sigma \cdot m - m)$ for all $\sigma \in G$ and some $m \in M$ such that $\beta(m) = n$. In particular, $\alpha \circ \phi \in B^1(G, M)$. Then

$$\alpha^1(\phi + B^1(G, P)) = \alpha \circ \phi + B^1(G, M) = B^1(G, M).$$

To prove $\supseteq$, let $\phi + B^1(G, P) \in \ker \alpha^1$. Then $\alpha \circ \phi \in B^1(G, M)$, that is $\alpha(\phi(\sigma)) = \sigma \cdot m - m$ for all $\sigma \in G$ and some $m \in M$. Note that

$$\delta(\beta(m)) = \psi + B^1(G, P),$$

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where $\psi(\sigma) = \alpha^{-1}(\sigma \cdot m - m)$. This implies that $\alpha(\psi(\sigma)) = \alpha(\phi(\sigma))$ for all $\sigma \in G$. Since α is injective, $\psi = \phi$. Therefore $\phi + B^1(G, P)$ belongs to the image of δ .

Finally, we prove that the sequence (21.1) is exact at $H^1(G, M)$, that is $\text{im } \alpha^1 = \ker \beta^1$. To prove $\subseteq$ note that

$$\beta^1(\alpha^1(\phi + B^1(G, P))) = \beta^1(\alpha \circ \phi + B^1(G, M)) = (\beta \circ \alpha) \circ \phi + B^1(G, N) = B^1(G, N).$$

Conversely, let $\phi + B^1(G, M) \in \ker \beta_1$. Then $\beta \circ \phi \in B^1(G, N)$. Thus there exists $n \in N$ such that $\beta(\phi(\sigma)) = \sigma \cdot n - n$ for all $\sigma \in G$. Since β is surjective, $n = \beta(m)$ for some $m \in M$. Hence

$$\beta(\phi(\sigma)) = \sigma \cdot n - n = \sigma \cdot \beta(m) - \beta(m) = \beta(\sigma \cdot m - m).$$

For each $\sigma \in G$, $\phi(\sigma) - (\sigma \cdot m - m) \in \ker \beta = \text{im } \alpha$. and therefore $\phi(\sigma) - (\sigma \cdot m - m) = \alpha(\rho_\sigma)$. This defines a map $\rho: G \rightarrow P$, $\sigma \mapsto \rho_\sigma$. We claim that $\rho \in Z^1(G, P)$. If $\sigma, \tau \in G$, then

$$\begin{aligned} \alpha(\rho_{\sigma\tau}) &= \phi(\sigma\tau) - ((\sigma\tau) \cdot m - m) \\ &= \phi(\sigma) + \sigma \cdot \phi(\tau) - (\sigma \cdot (\tau \cdot m - m) + \sigma \cdot m - m) \\ &= \alpha(\rho_\sigma) + \sigma \cdot \alpha(\rho_\tau). \end{aligned}$$

Since α is injective, it follows that $\rho \in Z^1(G, P)$. Now

$$\alpha_1(\rho + B^1(G, P)) = \alpha \circ \rho + B^1(G, M) = \phi + B^1(G, M). \quad \square$$

21.14. THEOREM. Let G be a finite group and M be a G -module. Then

$$|G|H^1(G, M) = \{0\}.$$

PROOF. Let $n = |G|$. It is enough to prove that if $\phi \in Z^1(G, M)$, then $n\phi \in B^1(G, M)$. Let $\phi \in Z^1(G, M)$ and $\sigma \in G$. Then

$$\phi(\sigma\tau) = \phi(\sigma) + \sigma \cdot \phi(\tau)$$

for all $\tau \in G$. Summing over all possible $\tau \in G$, we obtain that

$$(21.2) \quad \sum_{\tau \in G} \phi(\tau) = \sum_{\tau \in G} \phi(\sigma\tau) = \sum_{\tau \in G} \sigma \cdot \phi(\tau) + \sum_{\tau \in G} \phi(\sigma) = n\phi(\sigma).$$

Let $m = -\sum_{\tau \in G} \phi(\tau) \in M$. Then (21.2) can be rewritten as

$$-m = \sum_{\tau \in G} \phi(\tau) = \sigma \cdot \sum_{\tau \in G} \phi(\tau) + n\phi(\sigma) = -\sigma \cdot m + n\phi(\sigma).$$

Thus $n\phi(\sigma) = \sigma \cdot m - m$ and hence $n\phi \in B^1(G, M)$. $\square$

21.15. EXERCISE. Let G be a finite group and M be a finite G -module of size coprime to $|G|$. Prove that $H^1(G, M) = \{0\}$.

21.16. EXERCISE. Let G be a finite group and M be a finitely generated G -module. Prove that $H^1(G, M)$ is finite.

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